

Book II of Euclid's Elements

Formal Reconstruction — Technical Documentation

Jurek Różański

Military University of Technology, Warsaw, Poland

CGJTEAMLAB

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Document Status

This volume is the technical documentation on formal reconstruction of Book II of Euclid's *Elements*. It records the mathematical architecture of the reconstruction, the Lean formalization, the dependency structure of individual propositions, the reusable rectangle and scissors-congruence infrastructure, and the relation between the formal development and the classical sources.

The documentation accompanies the formal development. The Lean source is authoritative for theorem statements, hypotheses, dependencies, and proof status.

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Chapter 1

Proposition II.1

1.1 Introduction

Proposition II.1 opens Book II with a change of mathematical language. Book I has developed a synthetic theory of congruence, parallels, parallelograms, squares, and equal content. Book II begins to combine these ingredients in a systematic calculus of rectangles. A modern reader immediately recognizes in II.1 the pattern of a distributive law. That recognition is useful, but it must be stated carefully: Euclid does not introduce variables for numerical lengths, does not define multiplication of lengths, and does not prove an identity in a number field.

The primitive object in the proposition is a geometric figure: a rectangle “contained by” two straight lines. When one of the two lines is divided into parts, the large rectangle is divided into smaller rectangles having the same other side. The equality asserted by the proposition is therefore first of all a statement about decomposition of figures.

This is a natural point at which to speak of the beginning of the *geometrical algebra* characteristic of Book II. The phrase is an interpretive description of a stable pattern of geometric operations, not a claim that Euclid was secretly manipulating modern symbolic formulas. In the present formal reconstruction this distinction is especially sharp: the code contains rectangles, segment congruence, betweenness, parallelograms, right angles, and scissors relations, but no multiplication operation on segment lengths and no numerical area function.

The formal development separates three levels:

1. a concrete rectangle and its geometric cuts;
2. the Euclidean phrase “rectangle contained by” two given segments;
3. independence of the particular rectangle chosen to represent that phrase.

The final theorem is stronger than a bare equal-content statement. Its conclusion is the exact relation

```
HilbertScissorsEq
```

for the whole rectangle and the formal sum of its parts.

1.2 Euclid’s “Rectangle Contained By”

1.2.1 The original geometric meaning

Book II begins with the terminology that a rectangular parallelogram is “contained by” the two straight lines which contain one of its right angles. Thus the expression does not initially denote a product of two numbers. It names a rectangle by two adjacent sides.

Suppose a rectangle has adjacent sides congruent to two given segments PQ and RS . Euclid’s phrase

the rectangle contained by PQ, RS

should be read geometrically: choose a right-angled parallelogram whose two adjacent sides have those magnitudes. The rectangle itself remains a plane figure.

For exposition only, we will occasionally write

$$\text{Rect}(PQ, RS).$$

This is documentary shorthand. It is not a Lean definition, it is not a number-valued function, and the comma is not multiplication.

1.2.2 Why this already looks algebraic

If BC is divided into consecutive parts BM , MN , and NC , II.1 says in this shorthand

$$\text{Rect}(BC, PQ) = \text{Rect}(BM, PQ) + \text{Rect}(MN, PQ) + \text{Rect}(NC, PQ).$$

A modern algebraist naturally sees the shape

$$(b_1 + b_2 + b_3)a = b_1a + b_2a + b_3a.$$

The formal proof, however, establishes the geometric statement on the first line, not the numerical identity on the second. The second formula is only a modern mnemonic for the combinatorial pattern of the dissection.

This is consistent with the modern synthetic reading of Euclid’s area theory. Hartshorne treats Book II as a collection of equal-content statements about figures and describes the resulting manipulation as “geometrical algebra.” The point is precisely that the manipulation precedes any assignment of numerical areas to the figures.

1.3 The Synthetic Rectangle Layer

1.3.1 Rectangles

The reconstruction introduces a small reusable rectangle API in `HilbertRectangle.lean`. A rectangle is a parallelogram with one specified right angle:

```
def IsRectangle
  (A B C D : Geo.Point) : Prop :=
  IsParallelogram Geo A B C D /\
  HilbertRightAngle Geo A B C
```

The parallelogram structure supplies the opposite-side relations, while the right-angle component distinguishes rectangles from arbitrary parallelograms. The remaining right angles can then be propagated synthetically.

1.3.2 The formal meaning of “contained by”

The Euclidean phrase is represented by the predicate

```
def IsRectangleContainedBy
  (A B C D P Q R S : Geo.Point) : Prop :=
  IsRectangle Geo A B C D /\
  Geo.Congruent A B P Q /\
  Geo.Congruent B C R S
```

Thus

```
IsRectangleContainedBy Geo A B C D P Q R S
```

means exactly that $ABCD$ is a concrete rectangle, $AB \cong PQ$, and $BC \cong RS$. Nothing in this definition assigns a number to PQ or RS . The two given segments specify the two adjacent geometric magnitudes of a representative rectangle.

The layer also proves existence of such a representative and, more importantly for the semantics of Book II, uniqueness of its scissors content. The theorem

```
rectangle_contained_by_unique
```

says that any two rectangles contained by the same ordered pair of segments satisfy the exact scissors relation. Consequently Euclid’s definite description “the rectangle contained by PQ, RS ” becomes well-defined at the scissors level even though many different concrete quadrilaterals can represent it.

1.3.3 No segment multiplication and no numerical area

There are two tempting modernizations that the present reconstruction explicitly avoids.

First, we do not define a product of segment lengths $|PQ| |RS|$. The pair (PQ, RS) remains a pair of geometric segments, and the rectangle is related to that pair by congruence of adjacent sides.

Second, we do not map a rectangle to a real number called its area. The content of the proposition is expressed through relations between finite formal sums of triangles. This permits addition, dissection, transport, and comparison of figures at the synthetic level.

This is also conceptually distinct from Hilbert’s later arithmetic of segments, where arithmetic operations are introduced as an additional construction. No such field or segment-arithmetic layer is imported into II.1.

1.4 Exact Scissors Equality and Equal Content

The project uses two related but different content relations. They should not be conflated.

1.4.1 Exact scissors equality

`HilbertScissorsEq` records an exact finite cut-and-paste relation between formal scissors terms. In II.1 it is produced by explicit splits of the rectangle into smaller rectangles and by normalization of the triangulated parallelogram terms. Schematically,

$$P \simeq_{\text{sc}} Q + R$$

means that the two sides can be matched by the exact scissors machinery already developed in the project.

The rectangle transport theorem is already available in this stronger form:

```
rectangle_transport_scissorsEq
```

for rectangles with congruent corresponding adjacent sides.

1.4.2 The more general equal-content relation

The broader project notion corresponding to equal content is

```
HilbertScissorsEquicomplementable
```

It allows suitable complementary terms to be adjoined before exact scissors equality is established. Thus it is designed to capture a more general synthetic equal-content relation.

The theorem

```
rectangle_transport
```

is obtained by weakening the exact transport theorem from `HilbertScissorsEq` to equicomplementability.

For II.1 no such weakening is necessary. The final theorem proves `HilbertScissorsEq` itself. Therefore II.1 certainly yields an equal-content consequence, but its formal statement retains the stronger fact that the displayed equality comes from an exact dissection.

1.5 The Classical and Formal Configurations

1.5.1 The binary split

The reusable core is a one-cut theorem. Let $B - M - C$ and let the corresponding point L divide the opposite side DE . The rectangle $BCED$ is cut into the rectangles $BMLD$ and $MCEL$:

$$BCED \simeq_{\text{sc}} BMLD + MCEL.$$

In Lean the theorem is

```
rectangle_split
```

and it returns both the rectangle structure of the two pieces and the exact scissors equality.

The formal configuration contains one additional point X . It satisfies $D - X - C$ and $M - X - L$. This is not decorative data: the inherited exact split lemma from the I.47 scissors infrastructure uses the intersection of the cut line with a diagonal. The classical rectangle picture suppresses this witness; the formal theorem makes it explicit.

1.5.2 The three-part split

The three-part form applies the binary split twice. First cut at $M - L$; then cut the remaining rectangle at $N - K$. The result is

$$\begin{aligned} BCED &\simeq_{\text{sc}} BMLD \\ &\quad + MNKL \\ &\quad + NCEK. \end{aligned}$$

The second split has its own diagonal-intersection witness Y , with $L - Y - C$ and $N - Y - K$.

Figure 1.1 shows both levels and includes the hidden witnesses X and Y required by the current Lean interface.

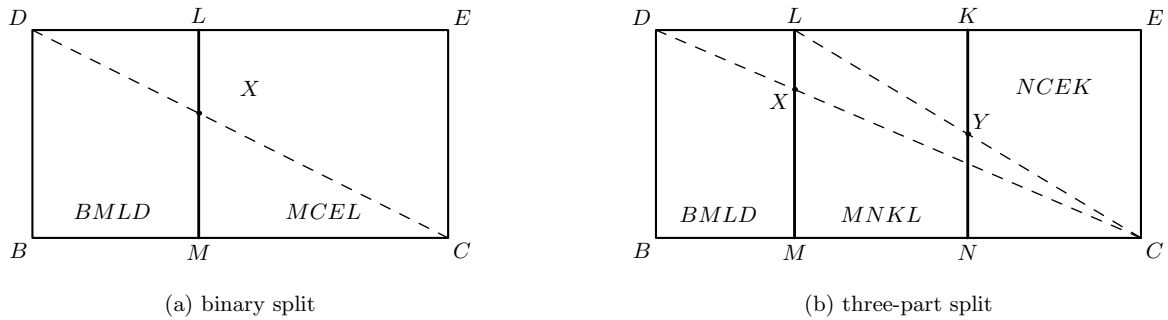


Figure 1.1: The exact rectangle splits underlying II.1. Panel (a) is the binary kernel: $B - M - C$ and the cut ML divide $BCED$ into $BMLD$ and $MCEL$; the dashed diagonal DC meets the cut at the formal witness X . Panel (b) applies the same construction twice: BC is divided into BM , MN , and NC , producing the three rectangles $BMLD$, $MNKL$, and $NCEK$. The dashed diagonals display the witnesses X and Y used by the two exact split calls.

The actual three-part theorem is

```
rectangle_split_three
```

Its proof is structurally transparent: obtain the first binary equality, split the remaining right-hand rectangle, add the unchanged left piece to that second equality, and compose by transitivity of `HilbertScissorsEq`.

1.5.3 Scope of the present formal statement

Euclid states II.1 for one line cut into an arbitrary finite number of parts. The current Lean file does not yet package an n -ary list-valued theorem. It proves the binary split and the explicit three-part form used in the standard II.1 configuration. The binary theorem is the reusable kernel from which a general finite version could later be obtained by iteration.

This is a deliberate statement-fidelity point. The documentation records Euclid’s general formulation, but the final formal statement displayed below is the representative-independent three-part theorem actually present in `Proposition2_1.lean`.

1.6 From a Concrete Diagram to Euclid’s Language

The geometric split alone produces concrete rectangles. Euclid II.1 is more abstract: it speaks of the rectangles *contained by* the whole segment and the several parts together with one fixed segment.

The theorem

```
rectangle_split_three_contained_by
```

bridges these levels. If $CE \cong PQ$, it packages the four concrete rectangles as

$$\begin{aligned} BCED & : \text{Rect}(BC, PQ), \\ BMLD & : \text{Rect}(BM, PQ), \\ MNKL & : \text{Rect}(MN, PQ), \\ NCEK & : \text{Rect}(NC, PQ). \end{aligned}$$

The opposite-side congruences of the parallelograms are what transport the fixed side PQ through the two cuts. The conclusion still contains the same exact scissors equality between the concrete quadrilateral terms.

At this stage the geometry corresponding to Euclid’s wording is visible, but the theorem still uses one particular drawn rectangle and its particular cut points. The final step removes this representational dependence.

1.7 Representative Independence

1.7.1 Why representatives matter formally

A phrase such as

$$\text{Rect}(BC, PQ)$$

does not identify four vertices. Lean, however, must manipulate an actual term such as

```
hilbertParallelogramTerm Geo U0 U1 U2 U3
```

with named vertices. The final theorem therefore accepts arbitrary representatives for the whole rectangle and for each of the three part rectangles.

The concrete dissection first proves

$$BCED \simeq_{\text{sc}} BMLD + MNKL + NCEK.$$

Then four calls to

```
rectangle_contained_by_unique
```

transport the whole and the three pieces to arbitrary representatives U , V , W , and Z . Compatibility of `HilbertScissorsEq` with formal addition transports the nested sum. Transitivity completes the chain

arbitrary whole representative
 $\Downarrow$
 concrete whole rectangle
 $\Downarrow$
 concrete three-part dissection
 $\Downarrow$
 arbitrary representatives of the three parts.

This is the formal content of representative independence. It is not merely a convenience for Lean. It explains why the Euclidean expression “the rectangle contained by two segments” can function as a stable geometric magnitude without first quotienting segments into numerical lengths or assigning numerical areas to rectangles.

1.8 Commutativity by Rectangle Rotation

The second algebra-like feature is symmetry in the two containing segments. For a numerical product one would write $ab = ba$. The synthetic proof does not appeal to a commutative multiplication law. Instead it rotates the presentation of a rectangle.

If $ABCD$ is contained by PQ, RS , then the cyclically reoriented rectangle $BCDA$ has adjacent sides BC and CD . The first is already congruent to RS ; the second is congruent to AB , hence to PQ . Thus the same geometric rectangle, with its vertices read from the next corner, is contained by RS, PQ .

The exact theorem available in the rectangle layer is

```
rectangle_contained_by_swap
```

It returns both the swapped `IsRectangleContainedBy` fact and a `HilbertScissorsEq` identifying the two quadrilateral presentations. The current rectangle API also packages the representative-independent consequence as

```
rectangle_contained_by_comm
```

so the commutativity needed by later Book II propositions is a theorem of rectangle geometry rather than an assumed law of segment multiplication.

Figure 1.2 shows the geometric mechanism. The operation is a cyclic change of the distinguished corner of the same rectangle; this exchanges the two adjacent side roles.

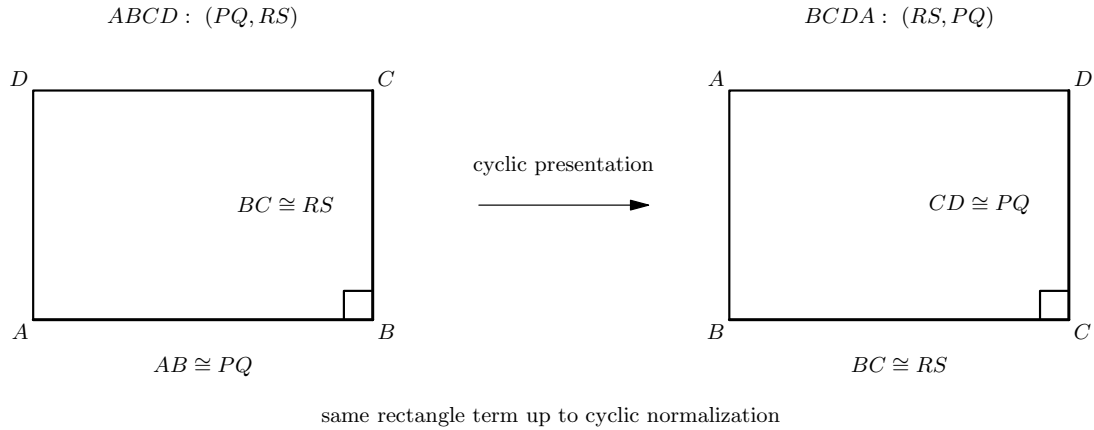


Figure 1.2: The geometric source of commutativity. On the left, $ABCD$ represents the rectangle contained by PQ, RS . Reading the same rectangle from the next vertex gives $BCDA$: now the first adjacent side represents RS and the second represents PQ . Exact scissors invariance under this cyclic presentation change yields the synthetic counterpart of exchanging the two factors.

This lemma is already operational in the next proposition. II.2 naturally writes rectangles in the order “whole side, part”, whereas the II.1 cut lemma produces them in the order “part, fixed side”. The implementation of II.2 uses the swap theorem to pass between these two presentations before applying the binary form of II.1.

1.9 Proof Architecture of the Reconstruction

The Lean proof is best read as five mathematical stages.

1.9.1 Step 1: exact binary dissection

```
rectangle_split_scissorsEq
rectangle_split_left
rectangle_split_right
rectangle_split
```

The inherited I.47 split theorem supplies the exact scissors decomposition. The two local rectangle helpers prove that each piece remains a rectangle by propagating right angles through the relevant parallelograms and along the same rays determined by betweenness.

1.9.2 Step 2: iterate the split

```
rectangle_split_three
```

The right-hand remainder of the first split is itself a rectangle, so the same binary theorem can be applied again. Addition and transitivity of exact scissors equality compose the two decompositions.

1.9.3 Step 3: attach the “contained by” data

```
rectangle_split_three_contained_by
```

Opposite-side congruence transports the fixed segment PQ through all three pieces. Each concrete piece is thereby promoted from `IsRectangle` to `IsRectangleContainedBy`.

1.9.4 Step 4: remove the choice of representatives

```
rectangle_contained_by_unique
euclid_proposition_2_1_three
```

The uniqueness theorem replaces the concrete whole and each concrete part by an arbitrary rectangle contained by the same pair of segments. This is the semantic step that makes the final statement faithful to Euclid’s definite phrase “the rectangle contained by.”

1.9.5 Step 5: expose the symmetric side order

```
rectangle_contained_by_swap
rectangle_contained_by_comm
```

Cyclic rotation of the rectangle exchanges the two containing segments. This is not needed to establish the displayed three-part split itself, but it is an important part of the Book II rectangle interface and is immediately useful in II.2 and later propositions.

1.10 Classical Proof and Formal Proof

1.10.1 Euclid’s proof pattern

In the classical three-part configuration, take one undivided segment A and a segment BC divided at D and E . Erect a perpendicular at one endpoint of BC , lay off on it a segment equal to A , complete the outer rectangle, and through the division points draw parallels to the perpendicular side. The outer rectangle is visibly partitioned into three rectangles. Each smaller rectangle is contained by A and one of the three consecutive parts of BC . Common Notion 2 then combines the parts into the whole equality.

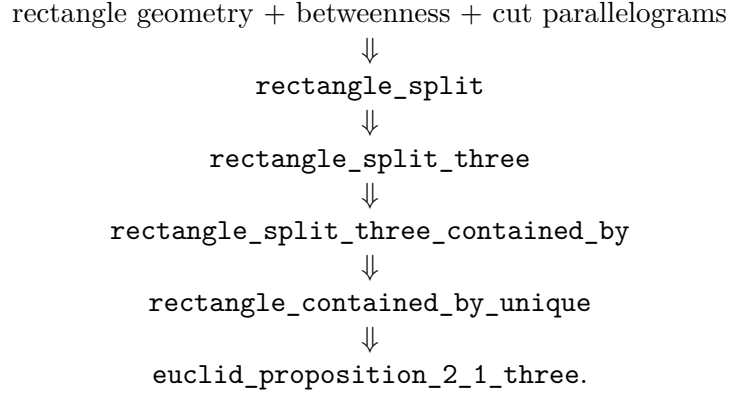
The classical DAG is therefore extremely small:

```

      divide one side
      ↓↓
draw the corresponding parallels
      ↓↓
      obtain three subrectangles
      ↓↓
whole rectangle = sum of the three subrectangles.
```

1.10.2 The formal DAG

The current reconstruction expands two things that the classical diagram treats as immediate: exact scissors bookkeeping and independence of the chosen rectangle representatives.



The proof is therefore not a mechanical transcription of Euclid. It preserves the geometric core, but makes the semantic and representational layers explicit.

1.11 Geometry, Content, and Representation

1.11.1 Geometry

The geometric data are substantial even though the proposition is visually simple. The current theorem receives the outer rectangle, strict betweenness relations on the divided side and its opposite, two diagonal-intersection witnesses, and four parallelogram facts for the two successive cuts. Right angles of the two resulting rectangles are not read from the picture; they are proved by right-angle transport and parallelogram geometry.

No coordinates are introduced. The cut points are controlled by **Between**, the side lengths by segment congruence, and the rectangular structure by parallels and synthetic right angles.

1.11.2 Content

Content enters in its strongest local form. The core statements are exact **HilbertScissorsEq** decompositions. The three-part theorem is composed from two exact binary splits by addition and transitivity. Representative transport is also exact scissors equality.

The more general equicomplementability relation is part of the surrounding Book I content theory, but II.1 does not need to weaken its conclusion to that level.

1.11.3 Representation

Representation is central to the formal meaning of the proposition. A rectangle contained by PQ, RS is not one canonical four-tuple of vertices. The predicate **IsRectangleContainedBy** describes a representative; **rectangle_contained_by_unique** proves that different representatives have the same exact scissors content.

There is a second representation issue inside the split itself. The inherited scissors theorem and the Book II statement use different cyclic orders for some parallelogram terms. The proof normalizes these by

```
parallelogram_term_rotateOne
```

before composing the scissors equalities. These cyclic changes are bookkeeping of a geometric representation, not new area facts.

1.12 Hidden Configuration Data

The classical drawing makes several facts appear automatic. Lean records them explicitly:

- $B - M - C$ for the first cut and $M - N - C$ for the second;
- $D - L - E$ and $L - K - E$ on the opposite side;
- $D - X - C$ together with $M - X - L$ for the first diagonal/cut intersection;
- $L - Y - C$ together with $N - Y - K$ for the second diagonal/cut intersection;
- explicit parallelogram witnesses for the left and right pieces of each split;
- congruence $CE \cong PQ$, which identifies the fixed side used in the “contained by” descriptions;
- four arbitrary rectangle representatives in the final theorem.

The points X and Y do not appear in Euclid’s statement and need not be emphasized in an elementary picture of II.1. They are nevertheless real data of the current exact-dissection API, inherited from the triangulated proof of the rectangle split.

1.13 Axiomatic Status

The public II.1 theorems assume

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

so II.1 is Euclidean in the current project API. Group IV is available through this typeclass and is consumed transitively by the parallelogram and parallel infrastructure on which the rectangle layer and inherited split theorem depend. The proposition itself does not introduce a new parallel axiom.

No continuity typeclass, Archimedean axiom, numerical area function, or segment field is introduced in `Proposition2_1.lean`. No proposition-local `axiom` or `sorry` occurs in the audited source. The modular builds for `HilbertRectangle` and `Proposition2_1` are recorded as clean at the present project checkpoint.

The content status is also worth separating from the later converse arguments of Book I. II.1 proceeds from geometry to an exact dissection. It does not infer metric geometry from an equal-content hypothesis, so it does not need a proper-part or De Zolt-type strictness argument.

1.14 Relationship with Book I and the Following Propositions

The exact split theorem reuses the scissors infrastructure established during I.47. In particular, the underlying theorem historically named

```
i47_square_split
```

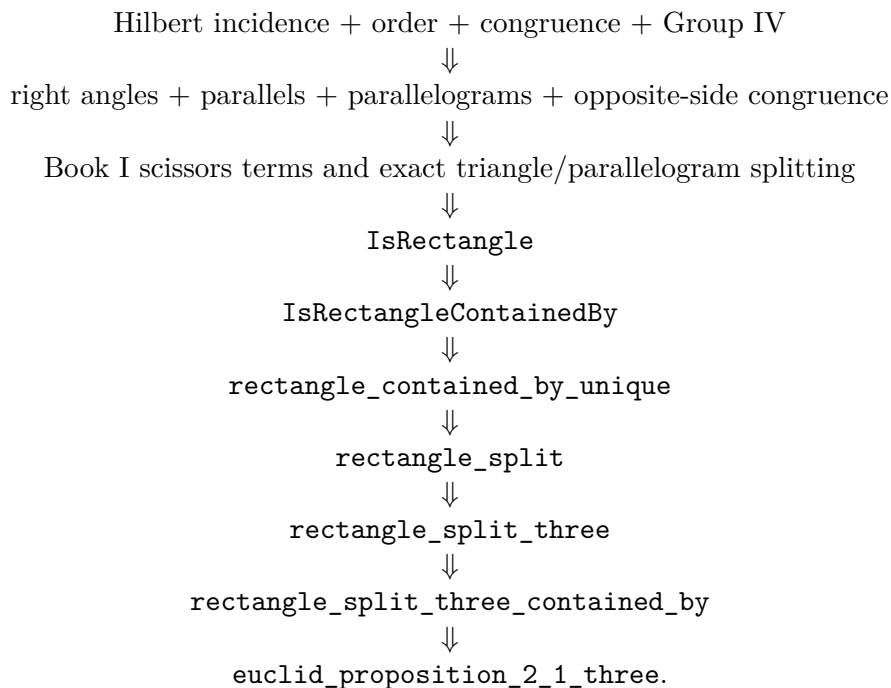
actually requires only parallelogram structure, and II.1 repackages it as a general rectangle split.

The rectangle layer also uses Book I results about parallelograms, opposite sides, congruent right angles, SAS/SSS transport, and the scissors representation of parallelograms. Thus Book II does not start a new foundational geometry; it starts a new way of organizing geometric magnitudes on top of the completed Book I infrastructure.

II.2 immediately imports II.1. Its formal proof illustrates why the symmetric rectangle API matters: Euclid's order of the two containing sides is opposite to the order produced by the binary cut theorem, so the representatives are swapped by rectangle rotation before II.1 is applied. II.3 continues the same rectangle calculus. In this sense II.1 is not only the first theorem of Book II but the API-level distributive kernel for the propositions that follow.

1.15 Dependency Summary

The actual formal dependency chain is



The symmetric side-order branch is

$$\begin{array}{c} \text{IsRectangleContainedBy}(PQ, RS) \\ \Downarrow \\ \text{cyclic rectangle rotation} \\ \Downarrow \\ \text{rectangle_contained_by_swap} \\ \Downarrow \\ \text{rectangle_contained_by_comm.} \end{array}$$

New geometric axiom in II.1:	no
New proposition-local axiom:	no
New rectangle abstraction:	yes, in <code>HilbertRectangle</code>
New reusable split helper:	yes
New representative-independence theorem:	yes
Exact scissors dissection used:	yes
General equicomplementability required by II.1:	no
Numerical area semantics:	no
Segment multiplication / field theory:	no
Group IV available/used transitively:	yes
Continuity assumption in the file:	no
Proposition-local <code>sorry/admit</code> :	no
Modular build:	clean

1.16 Conclusion

Proposition II.1 is the first clean instance in Book II of a product-like law being realized entirely by geometry. A segment is divided; a rectangle is cut along corresponding parallels; exact scissors equality identifies the whole with the sum of the pieces. The rectangle layer then removes the accidental choice of a concrete drawing, so Euclid’s phrase “the rectangle contained by” becomes a representative-independent synthetic magnitude.

The formalization also isolates the two structural laws that make the language look algebraic. Additivity comes from actual dissection. Commutativity comes from cyclic rotation of the rectangle. A square is the special case in which the two containing segments agree. None of these steps requires a product of segment lengths or a numerical area map.

Thus the “geometric algebra” of II.1 is best understood not as hidden modern symbolism but as a calculus of geometric magnitudes whose algebraic laws emerge from constructions, congruence, and finite decomposition.

The representative-independent three-part theorem at the end of the current formalization is:

```

theorem euclid_proposition_2_1_three
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]

  -- Concrete dissection diagram.
  (B C D E L M K N X Y P Q : Geo.Point)
  (hRect : IsRectangle Geo B C E D)
  (hCE_PQ : Geo.Congruent C E P Q)

  -- First cut.
```

```

(hBMC : Geo.Between B M C)
(hDLE : Geo.Between D L E)
(hDXC : Geo.Between D X C)
(hMXL : Geo.Between M X L)
(hLeftPar :
  IsParallelogram Geo L D B M)
(hRightPar :
  IsParallelogram Geo C E L M)

-- Second cut.
(hMNC : Geo.Between M N C)
(hLKE : Geo.Between L K E)
(hLYC : Geo.Between L Y C)
(hNYK : Geo.Between N Y K)
(hMiddlePar :
  IsParallelogram Geo K L M N)
(hFinalPar :
  IsParallelogram Geo C E K N)

-- Arbitrary representatives of the four rectangles.
(U0 U1 U2 U3 : Geo.Point)
(V0 V1 V2 V3 : Geo.Point)
(W0 W1 W2 W3 : Geo.Point)
(Z0 Z1 Z2 Z3 : Geo.Point)

(hWhole :
  IsRectangleContainedBy Geo
    U0 U1 U2 U3 B C P Q)

(hPart1 :
  IsRectangleContainedBy Geo
    V0 V1 V2 V3 B M P Q)

(hPart2 :
  IsRectangleContainedBy Geo
    W0 W1 W2 W3 M N P Q)

(hPart3 :
  IsRectangleContainedBy Geo
    Z0 Z1 Z2 Z3 N C P Q) :
HilbertScissorsEq Geo
  (hilbertParallelogramTerm Geo U0 U1 U2 U3)
  (hilbertParallelogramTerm Geo V0 V1 V2 V3 +
    (hilbertParallelogramTerm Geo W0 W1 W2 W3 +
      hilbertParallelogramTerm Geo Z0 Z1 Z2 Z3)) := by

```

Chapter 2

Proposition II.2

2.1 Introduction

Proposition II.2 is the first place where the rectangle calculus established in II.1 is reused rather than developed further. The classical proposition is visually elementary: a square on a straight line is cut by a parallel through an arbitrary point of that line, and the two resulting rectangles fill the square. The formal reconstruction preserves exactly that geometric content, but its proof architecture is deliberately different from Euclid’s.

Euclid proves II.2 directly from a square construction and a parallel through the point of division. The current Lean theorem instead imports `Proposition2_1` and applies the representative-independent binary form of II.1. One additional operation is needed: Euclid names each part rectangle with the whole side first, whereas the II.1 cut theorem returns the divided part first. The proof therefore rotates each rectangle representative by one vertex, applies II.1, and then transports the result back to Euclid’s original order.

This makes II.2 a particularly clean test of the Book II API. Two algebra-like laws are already available, but neither is primitive algebra:

- additivity is realized by an exact rectangle dissection;
- exchange of the two containing segments is realized by cyclic rotation of a rectangle.

No multiplication of segment lengths and no numerical area function occurs in the theorem.

2.2 Euclid’s Statement and Classical Configuration

2.2.1 The statement

In Heath’s wording, Proposition II.2 states:

If a straight line be cut at random, the rectangle contained by the whole and both of the segments is equal to the square on the whole.

Here “both of the segments” means the two rectangles obtained by pairing the whole line with each of the two parts. If AB is cut at C , the geometric statement is therefore

$$\text{Rect}(AB, AC) + \text{Rect}(AB, CB) = \text{Sq}(AB).$$

The notation is documentary shorthand only. It does not introduce a number-valued rectangle function.

2.2.2 The classical diagram

Euclid takes a straight line AB cut at C , constructs the square $ADEB$ on AB by I.46, and through C draws CF parallel to the two opposite sides AD and BE by I.31 [1]. The square is thereby partitioned into two rectangles. One is contained by the whole AB and the segment AC ; the other is contained by the whole AB and the segment CB .

The current Lean configuration uses different letters:

classical role	Lean role
AB	BC
C	M
$ADEB$	$BCED$
CF	ML

Thus $B - M - C$, the outer figure $BCED$ is a square, and the cut ML divides it into the left and right rectangular pieces.

Figure 2.1 shows both the configured cut and the separate representation change used by the formal proof.

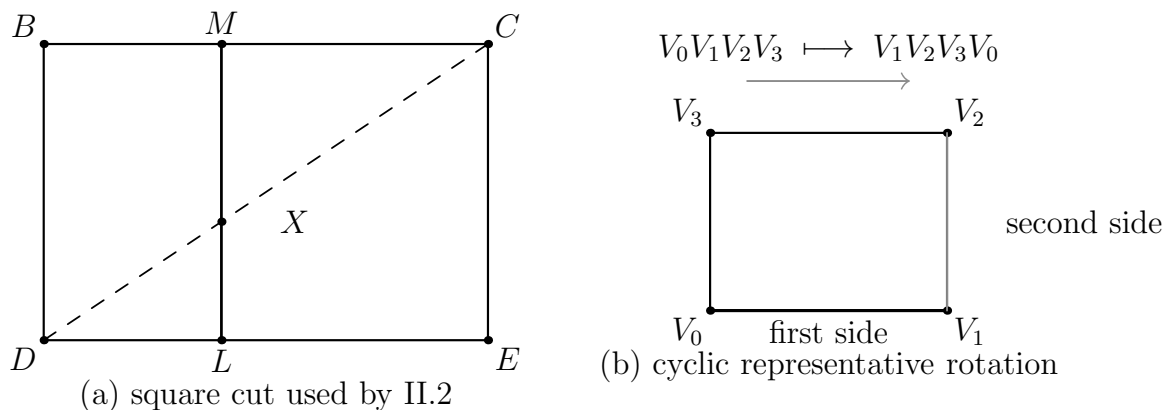


Figure 2.1: Two mechanisms used in the formal II.2 proof. Panel (a) is the instantiated square $BCED$ with $B - M - C$ and the cut ML ; the dashed diagonal DC meets the cut at the hidden witness X , so the formal data include both $D - X - C$ and $M - X - L$. Panel (b) shows the cyclic presentation change $V_0V_1V_2V_3 \mapsto V_1V_2V_3V_0$, which exchanges the two adjacent side roles while preserving exact scissors content.

2.3 Classical Proof

The classical proof has only four mathematical steps.

First, construct the square on the whole segment. Second, draw through the point of section a line parallel to the appropriate pair of sides of the square. Third, observe that this line partitions the square into two rectangles. Finally, identify those rectangles by Book II Definition 1: their adjacent sides are the whole segment together with the two parts of the divided segment.

In the standard lettering this gives

$$\text{Rect}(AB, AC) + \text{Rect}(AB, CB) = \text{Sq}(AB).$$

Euclid does not invoke II.1 in this proof. The local classical dependency graph is instead

I.46: construct the square on AB
 $\Downarrow$
 I.31: draw the parallel through C
 $\Downarrow$
 II.Def.1: identify the two rectangles
 $\Downarrow$
 the two pieces constitute the square.

This fact is important for documentation. The formal theorem proves the same geometric identity, but it is not a line-by-line formalization of Euclid's historical derivation.

2.4 The Geometric Identity

In the current Lean lettering the square side is BC and the cut point is M . The two Euclidean rectangles are therefore

$$\text{Rect}(BC, BM), \quad \text{Rect}(BC, MC),$$

and the theorem asserts the exact scissors relation

$$\text{Rect}(BC, BM) + \text{Rect}(BC, MC) \simeq_{\text{sc}} \text{Sq}(BC).$$

A modern algebraic shadow is obtained by writing $x = y + z$:

$$x^2 = xy + xz.$$

This is useful as a mnemonic for the pattern, but it is not the object proved by Lean and it is not a claim that Euclid has defined multiplication on segment lengths. The formal terms remain rectangle representatives and their exact scissors relation.

2.5 Reusable Book II Infrastructure

II.2 imports

```
import CGJteamLab.Proposition2_1
```

and introduces no new rectangle abstraction. Its proof consumes four pieces of the existing Book II infrastructure.

2.5.1 A square as a rectangle contained by the same side twice

From

```
hSquare : IsSquare Geo B C E D
```

the theorem

```
square_is_rectangle_contained_by_side
```

produces

```
IsRectangleContainedBy Geo B C E D B C B C
```

so the concrete square is recognized synthetically as a rectangle contained by BC, BC .

2.5.2 The binary form of II.1

The reusable distributive kernel is

```
euclid_proposition_2_1_two
```

whose geometric content is

$$\text{Rect}(BC, PQ) \simeq_{\text{sc}} \text{Rect}(BM, PQ) + \text{Rect}(MC, PQ)$$

when $B - M - C$ and the corresponding rectangular cut data are supplied.

For II.2 the fixed segment is specialized to $PQ = BC$. Thus the right-hand pieces naturally arrive in the order

$$\text{Rect}(BM, BC), \quad \text{Rect}(MC, BC).$$

2.5.3 Exchanging the containing-side order

Euclid’s statement uses the opposite order:

$$\text{Rect}(BC, BM), \quad \text{Rect}(BC, MC).$$

The current proof does not treat this as “commutativity of multiplication.” It calls

```
rectangle_contained_by_swap
```

on each arbitrary representative.

The theorem rotates a rectangle

$$A B C D \mapsto B C D A,$$

returns the corresponding `IsRectangleContainedBy` fact with the two containing segments exchanged, and simultaneously returns a `HilbertScissorsEq` relating the two cyclic presentations.

The higher-level theorem `rectangle_contained_by_comm` belongs to the same rectangle API, but the public II.2 proof uses `rectangle_contained_by_swap` directly because its returned data are exactly the swapped representatives and exact scissors transports needed for the subsequent II.1 call.

2.6 Proof Architecture of the Reconstruction

The Lean proof has four stages.

2.6.1 Step 1: recognize the square

The square hypothesis is converted to the “contained by” language:

```
have hSquareContained :
  IsRectangleContainedBy Geo
    B C E D B C B C :=
square_is_rectangle_contained_by_side
  Geo B C E D hSquare
```

The rectangle structure and the side congruence $CE \cong BC$ are then extracted from this fact.

2.6.2 Step 2: rotate the two arbitrary part representatives

The two hypotheses supplied in Euclid’s order are

```
hPart1 :
  IsRectangleContainedBy Geo
    V0 V1 V2 V3 B C B M

hPart2 :
  IsRectangleContainedBy Geo
    W0 W1 W2 W3 B C M C
```

The current source then applies `rectangle_contained_by_swap` separately to `hPart1` and `hPart2`. Each call returns a swapped `IsRectangleContainedBy` fact and an exact scissors equality for the cyclically rotated representative. The resulting representatives are $V_1V_2V_3V_0$ and $W_1W_2W_3W_0$.

The mathematical result is that the rotated representatives now have the side orders required by the II.1 binary theorem.

2.6.3 Step 3: apply II.1 with fixed segment BC

The central call is

```
have hSplit :
  HilbertScissorsEq Geo
    (hilbertParallelogramTerm Geo B C E D)
    (hilbertParallelogramTerm Geo V1 V2 V3 V0 +
      hilbertParallelogramTerm Geo W1 W2 W3 W0) :=
euclid_proposition_2_1_two
  Geo
  B C D E L M X
  B C
  hRect
```

```

hCE_BC
hBMC
hDLE
hDXC
hMXL
hLeftPar
hRightPar
B C E D
V1 V2 V3 V0
W1 W2 W3 W0
hSquareContained
hPart1Swap
hPart2Swap

```

This is the whole mathematical core of the reconstruction:

$$\text{Sq}(BC) \simeq_{\text{sc}} \text{Rect}(BM, BC) + \text{Rect}(MC, BC).$$

2.6.4 Step 4: return to Euclid's ordering

The two exact rotation equalities are added:

```

have hPartsBack :
  HilbertScissorsEq Geo
    (hilbertParallelogramTerm Geo V1 V2 V3 V0 +
     hilbertParallelogramTerm Geo W1 W2 W3 W0)
    (hilbertParallelogramTerm Geo V0 V1 V2 V3 +
     hilbertParallelogramTerm Geo W0 W1 W2 W3) :=
HilbertScissorsEq.add
  (Geo := Geo)
  (HilbertScissorsEq.symm
   (Geo := Geo) hRotate1)
  (HilbertScissorsEq.symm
   (Geo := Geo) hRotate2)

```

Transitivity composes `hSplit` with `hPartsBack`, and the final symmetry reverses the result into the order used by the public theorem: sum of the two Euclidean rectangles on the left, square on the right.

2.7 Lean Formalization

The current public theorem is the following instantiated diagram theorem:

```

theorem euclid_proposition_2_2
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]

  (B C D E L M X : Geo.Point)

  (hSquare : IsSquare Geo B C E D)

  (hBMC : Geo.Between B M C)

```

```

(hDLE : Geo.Between D L E)
(hDXC : Geo.Between D X C)
(hMXL : Geo.Between M X L)
(hLeftPar :
  IsParallelogram Geo L D B M)
(hRightPar :
  IsParallelogram Geo C E L M)

(V0 V1 V2 V3 : Geo.Point)
(W0 W1 W2 W3 : Geo.Point)

(hPart1 :
  IsRectangleContainedBy Geo
    V0 V1 V2 V3 B C B M)

(hPart2 :
  IsRectangleContainedBy Geo
    W0 W1 W2 W3 B C M C) :
HilbertScissorsEq Geo
  (hilbertParallelogramTerm Geo V0 V1 V2 V3 +
   hilbertParallelogramTerm Geo W0 W1 W2 W3)
  (hilbertParallelogramTerm Geo B C E D) := by

```

Three aspects of the signature are mathematically significant.

First, the theorem assumes an instantiated square and cut diagram. It is not an existence theorem constructing the square and the parallel cut from only a segment and a cut point. Euclid performs those constructions in the classical proof; the present public Lean theorem starts after they have been supplied.

Second, the two part rectangles are arbitrary representatives satisfying `IsRectangleContainedBy`. Their exact placement in the plane is not fixed by the statement.

Third, the conclusion is the exact relation `HilbertScissorsEq`. It is not merely a conclusion in the more general `HilbertScissorsEquicomplementable` relation. II.2 remains at the exact-dissection level because its proof is ultimately the binary exact split of II.1 plus exact representation transports.

2.8 Relationship with Proposition II.1

The contrast between the classical and formal dependency graphs is especially clear here.

Euclid's II.2 is proved directly from the square and the parallel cut. The current Lean file instead begins with

```
import CGJteamLab.Proposition2_1
```

and the public proof calls

```
euclid_proposition_2_1_two
```

once.

Thus II.2 is an explicit formal corollary of the binary II.1 API. No new dissection lemma is proved. The only additional mathematical work is conversion between the two possible orders of the containing segments.

This reuse also explains why the symmetric rectangle interface added at II.1 is not cosmetic. Without the side-order transport, the II.1 theorem would produce perfectly correct rectangles whose formal orientation does not match the language of II.2.

2.9 Relationship with Book I Infrastructure

The direct import is II.1, not a Book I proposition. Nevertheless the inherited chain underneath II.1 contains the Book I geometry on which the rectangle calculus rests.

The exact binary split is built over the scissors decomposition infrastructure developed around I.47. Rectangle structure uses Euclidean parallelogram theory and synthetic right angles; the square hypothesis uses the square infrastructure established in Book I. Consequently the formal proof of II.2 does not replay I.31 and I.46 as Euclid does. Those classical constructions have been absorbed into the established geometric API and into the hypotheses of the instantiated theorem.

2.10 Formalization Notes and Hidden Geometric Data

The visible square cut hides several obligations that are explicit parameters of the current theorem:

- $B - M - C$, fixing the orientation of the cut on the square side;
- $D - L - E$, locating the opposite endpoint of the cut;
- $D - X - C$ and $M - X - L$, recording the intersection X of the cut with the diagonal used by the inherited exact scissors split;
- `IsParallelogram Geo L D B M` for the left piece;
- `IsParallelogram Geo C E L M` for the right piece;
- two arbitrary representatives of the Euclidean part rectangles;
- two cyclic representative changes, each accompanied by an exact scissors equality.

The point X has no role in the elementary verbal statement. It belongs to the current exact-dissection interface inherited from the triangulated scissors proof. It is therefore documented as hidden formal geometry rather than presented as part of Euclid's original construction.

There is also an important negative observation. No `SameRay`, `SameSide`, or `OppositeSide` obligation appears in `Proposition2_2.lean` itself. The corresponding low-level geometry has already been discharged inside the reusable rectangle and II.1 theorems.

2.11 Geometry, Content, Representation, and Later Arithmetic

2.11.1 Geometry

The geometric layer consists of a square, betweenness on two opposite sides, the diagonal/cut witness, and two parallelogram pieces. Congruence of the square side with itself identifies the square as a rectangle contained by BC, BC .

2.11.2 Content / scissors

The conclusion and every transport used in the proof are exact `HilbertScissorsEq` relations. The proof uses symmetry, transitivity, and addition of exact scissors equalities. It does not weaken to the more general equicomplementability relation.

2.11.3 Representation

Representation is the distinctive issue in II.2. The theorem accepts arbitrary rectangles for

$$\text{Rect}(BC, BM) \quad \text{and} \quad \text{Rect}(BC, MC).$$

The binary II.1 theorem wants the factors in the reverse order. Cyclic rotation of each representative supplies the required conversion while preserving exact scissors content.

2.11.4 Later arithmetic / numerical area

Neither layer is used. In particular, the proof does not define a segment product, does not invoke Hilbert’s later segment field, and does not assign real areas to the square or rectangles.

2.12 Axiomatic and Semantic Status

The public theorem assumes

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

and therefore works in the same Euclidean project layer as II.1. No new proposition-specific geometric axiom or content axiom is declared in `Proposition2_2.lean`. Inspection of the current 162-line source shows one theorem declaration and no proposition-local `axiom`, `sorry`, or `admit`.

This should not be described as “axiom-free.” The theorem inherits the axioms and typeclass assumptions of the established Hilbert/Euclidean/scissors library transitively. No current `#print axioms` output was supplied with the source used for this documentation pass, so the transitive axiom list is not reproduced here.

The implementation checkpoint records that the II.2 Lean build had already been checked during theorem development. This documentation workspace does not contain the complete Lean project, so the Lean build is not re-run here; the LaTeX and Asymptote checks are reported separately.

2.13 Hartshorne’s Semantic Control

Hartshorne places the end of Book I and all of Book II inside Euclid’s theory of area, interpreted synthetically as equal content rather than as an unexplained numerical area function [5]. He also describes Book II as part of an “algebra of areas”: rectangles can play a product-like role because Euclid has no operation that multiplies two line segments to produce another segment.

II.2 fits that interpretation exactly, provided the levels are kept separate. The square on the whole and the two rectangles are geometric figures; their equality is a content statement; the modern identity $x^2 = xy + xz$ is only an interpretive shadow.

The current Lean theorem is locally stronger than the weakest equal-content reading because its configured proof stays at exact scissors equality. That strength comes from the explicit binary dissection inherited from II.1, not from a general principle identifying every equal-content pair with an exact dissection.

2.14 Hilbert’s Layering

Hilbert supplies a second semantic guardrail. His theory separates synthetic plane content from the later arithmetic of segments. Segment multiplication is introduced only in a later arithmetic construction, with a fixed unit segment and parallel geometry; it is not primitive rectangle notation.

The II.2 proof remains entirely on the earlier side of this boundary:

square and rectangle geometry
 $\Downarrow$
 exact scissors dissection
 $\Downarrow$
 representative rotation and transport.

The apparent commutativity required by the formula $xy = yx$ is realized geometrically by `rectangle_contained_by_swap`, not imported from Hilbert’s later segment arithmetic.

2.15 Dependency Summary

The actual Lean dependency chain is

Hilbert incidence + Euclidean plane layer
 $\Downarrow$
 Book I right-angle, square, parallelogram, and scissors infrastructure
 $\Downarrow$
`HilbertRectangle : IsRectangleContainedBy, rectangle_contained_by_swap`
 $\Downarrow$
 Proposition II.1 binary API
 $\Downarrow$
`euclid_proposition_2_1_two`
 $\Downarrow$
 two representative rotations + exact scissors addition/transitivity
 $\Downarrow$
`euclid_proposition_2_2`.

The classical local chain is different:

$$\text{I.46} \longrightarrow \text{I.31} \longrightarrow \text{square cut into two rectangles} \longrightarrow \text{II.2.}$$

New geometric axiom in II.2:	no
New content axiom in II.2:	no
New proposition-local helper theorem:	no
New reusable helper theorem:	no
New <code>HilbertRectangle</code> theorem introduced by II.2:	no
New Interface / Book Zero / Scissors theorem:	no
Reuses an earlier Book II proposition:	yes, II.1
Direct side-order transport used:	yes, <code>rectangle_contained_by_swap</code>
Exact <code>HilbertScissorsEq</code> :	yes
General equicomplementability required:	no
Numerical area:	no
Segment multiplication:	no
Public Lean theorem build:	previously checked; not re-run here
Full relevant Lean build:	previously checked; not re-run here
Proposition-local <code>sorry/admit</code> :	no

2.16 Conclusion

Proposition II.2 is mathematically a square cut into two rectangles, but formally it does more than repeat the picture. It demonstrates that the II.1 rectangle calculus is already reusable: the binary dissection theorem supplies additivity, and cyclic rectangle rotation reconciles the two possible orders of the containing sides.

This is also the first sharp divergence between Euclid’s local proof graph and the formal Book II graph. Euclid proves II.2 directly from I.46 and I.31. Lean uses II.1 as a library theorem and turns II.2 into a short transport-and-reuse argument. The theorem is therefore not a literal transcription of the classical proof, but it remains entirely synthetic and geometric.

The result can be remembered by the modern shadow

$$x^2 = xy + xz, \quad x = y + z,$$

but the formal content is more precise: arbitrary representatives of the two rectangles contained by the whole side and the two parts are exactly scissors-equal, in sum, to the concrete square on the whole side. No numerical area and no multiplication of segment lengths is required.

Chapter 3

Proposition II.3

3.1 Introduction

Proposition II.3 is the second immediate reuse of the rectangle calculus established in Proposition II.1. Its classical form is again a statement about a straight line cut at an arbitrary point, but now one of the two resulting segments plays a distinguished role. The rectangle contained by the whole line and that segment is decomposed into a rectangle contained by the two parts and the square on the distinguished segment.

The current Lean reconstruction makes this structure unusually transparent. Its only direct import is the II.1 module:

```
import CGJteamLab.Proposition2_1
```

The proof first recognizes the supplied square on MC as a rectangle contained by MC, MC , and then applies the binary form of II.1 with the fixed segment specialized to MC . No new dissection theorem, side-order swap, or proposition-local helper theorem is needed.

This is an important contrast with II.2. There the Euclidean order of the two containing segments did not match the order produced by the binary II.1 theorem, so cyclic rectangle rotation was required. In II.3 the order already matches:

$$\text{Rect}(BC, MC) \simeq_{\text{sc}} \text{Rect}(BM, MC) + \text{Rect}(MC, MC).$$

The square on MC is then simply a chosen representative of the final $\text{Rect}(MC, MC)$ term.

Thus II.3 is not formalized through multiplication of segment lengths or through a numerical area function. Its apparently algebraic shape is realized by an exact geometric cut together with representative independence in the rectangle API.

3.2 Euclid's Statement and Classical Configuration

3.2.1 The statement

In Heath's wording, Proposition II.3 states:

If a straight line be cut at random, the rectangle contained by the whole and one of

the segments is equal to the rectangle contained by the segments and the square on the aforesaid segment.

If AB is cut at C and the distinguished part is BC , the geometric identity is

$$\text{Rect}(AB, BC) = \text{Rect}(AC, CB) + \text{Sq}(BC).$$

As throughout the Book II documentation, Rect and Sq are expository shorthand for geometric figures and their content. They are not number-valued functions.

3.2.2 The classical diagram

Euclid constructs the square $CDEB$ on the distinguished segment CB by Proposition I.46. The side ED is produced, and through A a line is drawn parallel to the corresponding sides of the square by Proposition I.31 [1]. The resulting larger rectangle has adjacent side magnitudes AB and BC . The line through C separates it into a rectangle with side magnitudes AC, CB and the square on CB .

The current Lean lettering is related to the classical roles by

classical role	Lean role
AB	BC
C	M
AC	BM
CB	MC

so the formal cut relation is $B - M - C$ and the distinguished segment is MC . The configured outer rectangle is $BCED$, with $CE \cong MC$, while the cut through M meets the opposite side at L .

Figure 3.1 shows the geometry used by the reconstruction. The rectangle $BCED$ is drawn below the segment BC . Its right-hand piece has width MC and height congruent to MC , so it realizes the square-like $\text{Rect}(MC, MC)$ part of the dissection. The supplied square $MCFG$ is drawn on the opposite side of MC to emphasize a formal point: the conclusion may use a square representative different from the concrete piece created by the cut.

3.3 Classical Proof

Let AB be cut at C , and choose BC as the segment named in the proposition. Construct the square $CDEB$ on CB by I.46. Produce one side of the square and through A draw a line parallel to the opposite sides by I.31. This forms a larger rectangle whose adjacent side magnitudes are AB and BC .

The line through C partitions that larger rectangle into two pieces. The first has adjacent side magnitudes AC and CB , and the second is precisely the square on CB . Hence

$$\text{Rect}(AB, BC) = \text{Rect}(AC, CB) + \text{Sq}(BC).$$

This is Euclid's entire content argument: the whole rectangle is constituted by the two displayed subfigures.

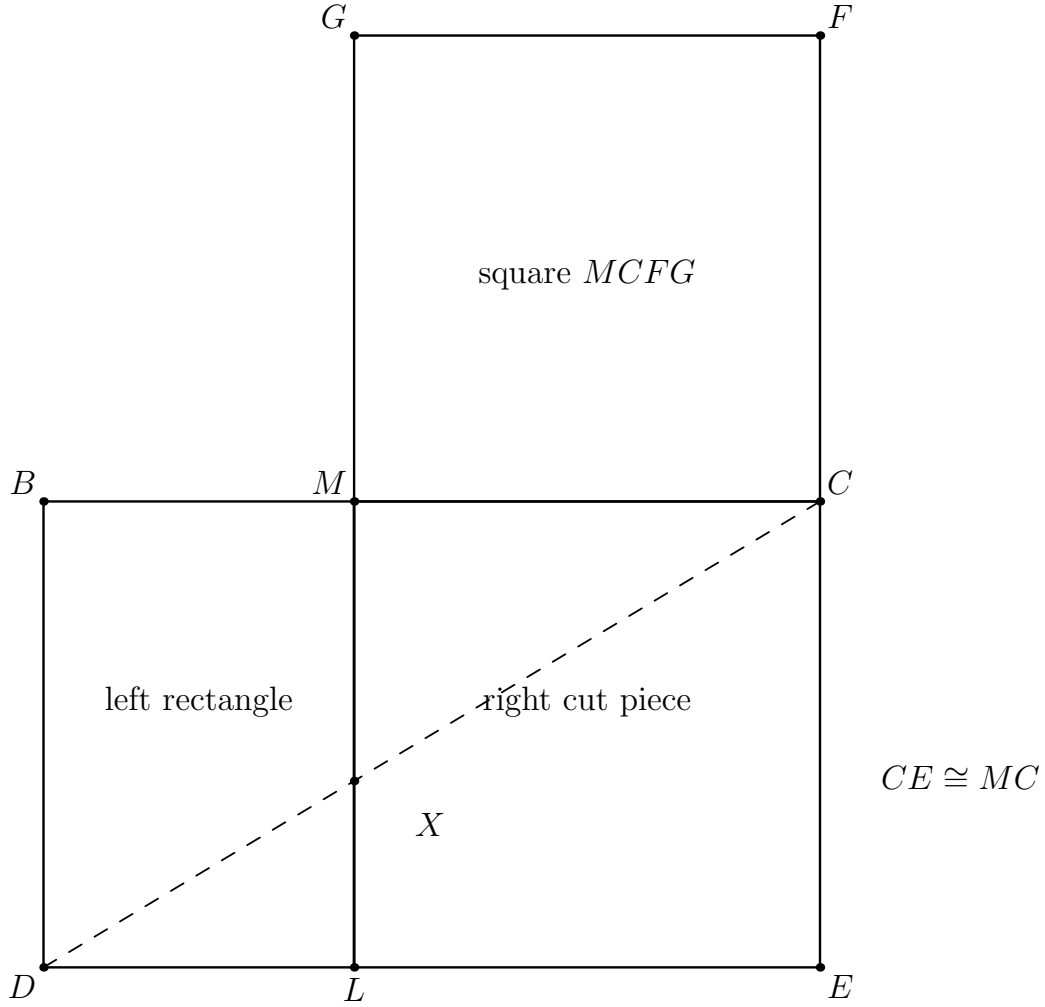


Figure 3.1: The geometric mechanism of II.3. The outer rectangle $BCED$ has height $CE \cong MC$ and is split by ML at the point $B - M - C$. The dashed diagonal DC meets the cut at the hidden witness X . The right-hand cut piece represents $\text{Rect}(MC, MC)$; the separately drawn square $MCFG$ is the square representative used in the public theorem.

The local classical dependency graph is therefore

$$\begin{array}{c}
 \text{I.46: construct the square on } CB \\
 \Downarrow \\
 \text{I.31: draw the required parallel through } A \\
 \Downarrow \\
 \text{Book II Definition 1: identify the rectangles} \\
 \Downarrow \\
 \text{whole rectangle} = \text{rectangle} + \text{square} \\
 \Downarrow \\
 \text{II.3.}
 \end{array}$$

Euclid does not cite Proposition II.1 in this proof. The current Lean proof, by contrast, is explicitly organized as a corollary of the reusable II.1 binary theorem. The classical and formal dependency graphs therefore prove the same geometric identity through different library organizations.

3.4 The Geometric Identity

In the current Lean lettering, $B - M - C$. The theorem fixes the segment MC and applies it to the whole BC and the two parts BM, MC . Its exact geometric content is

$$\text{Rect}(BC, MC) \simeq_{\text{sc}} \text{Rect}(BM, MC) + \text{Sq}(MC).$$

A modern algebraic shadow is obtained by writing

$$x = y + z,$$

where x corresponds to BC , y to BM , and z to MC . Then the pattern is

$$xz = yz + z^2,$$

or equivalently

$$(y + z)z = yz + z^2.$$

This formula is useful only as a mnemonic. The Lean proof never forms the products xz or yz as segment-arithmetic objects. The actual objects are rectangle representatives and a square, related by exact scissors equality.

3.5 Reusable Book II Infrastructure

The current file begins with

```
import CGJteamLab.Proposition2_1
```

and introduces no new rectangle abstraction.

3.5.1 The binary form of II.1

The reusable theorem is

```
euclid_proposition_2_1_two
```

with geometric content

$$\text{Rect}(BC, PQ) \simeq_{\text{sc}} \text{Rect}(BM, PQ) + \text{Rect}(MC, PQ)$$

whenever $B - M - C$ and the required rectangular cut data are supplied.

For II.3 the fixed segment is specialized directly to

$$PQ = MC.$$

Therefore II.1 returns

$$\text{Rect}(BC, MC) \simeq_{\text{sc}} \text{Rect}(BM, MC) + \text{Rect}(MC, MC),$$

which already has exactly the containing-side order required by the public II.3 theorem.

3.5.2 A square as $\text{Rect}(MC, MC)$

The square hypothesis

```
hSquare : IsSquare Geo M C F G
```

is converted by

```
square_is_rectangle_contained_by_side
```

into

```
IsRectangleContainedBy Geo M C F G M C M C
```

so the supplied square is a concrete representative of the final $\text{Rect}(MC, MC)$ term.

3.5.3 What II.3 does not use

Unlike II.2, the current proof does not call

```
rectangle_contained_by_swap  
rectangle_contained_by_comm
```

The absence is mathematically informative. The binary II.1 theorem already places BM and MC first and the fixed segment MC second, exactly as II.3 needs. No geometric analogue of exchanging factors is required.

3.6 Proof Architecture of the Reconstruction

The public proof has only two mathematical stages.

3.6.1 Step 1: recognize the square

The first local fact is

```
have hSquareContained :  
  IsRectangleContainedBy Geo  
    M C F G M C M C :=  
  square_is_rectangle_contained_by_side  
    Geo M C F G hSquare
```

This moves from the square-specific geometric predicate to the general “rectangle contained by” language used by II.1.

3.6.2 Step 2: instantiate II.1 with the fixed segment MC

The remainder of the proof is one direct call:

```
exact
  euclid_proposition_2_1_two
    Geo
    B C D E L M X
    M C
    hRect
    hCE_MC
    hBMC
    hDLE
    hDXC
    hMXL
    hLeftPar
    hRightPar
    U0 U1 U2 U3
    V0 V1 V2 V3
    M C F G
    hWhole
    hProduct
    hSquareContained
```

No scissors transitivity, addition, symmetry, or representative rotation is performed locally in II.3. Those mechanisms have already been packaged inside the representative-independent binary theorem of II.1.

This makes the formal DAG exceptionally short:

$$\begin{array}{c}
 \text{square } MCFG \\
 \Downarrow \\
 \text{square_is_rectangle_contained_by_side} \\
 \Downarrow \\
 \text{Rect}(MC, MC) \text{ representative} \\
 \Downarrow \\
 \text{euclid_proposition_2_1_two with } PQ = MC \\
 \Downarrow \\
 \text{euclid_proposition_2_3.}
 \end{array}$$

3.7 Lean Formalization

The current public theorem is:

```
theorem euclid_proposition_2_3
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]

  (B C D E L M X : Geo.Point)
  (hRect : IsRectangle Geo B C E D)
  (hCE_MC : Geo.Congruent C E M C)

  (hBMC : Geo.Between B M C)
```

```

(hDLE : Geo.Between D L E)
(hDXC : Geo.Between D X C)
(hMXL : Geo.Between M X L)
(hLeftPar :
  IsParallelogram Geo L D B M)
(hRightPar :
  IsParallelogram Geo C E L M)

(F G : Geo.Point)
(hSquare : IsSquare Geo M C F G)

(U0 U1 U2 U3 : Geo.Point)
(V0 V1 V2 V3 : Geo.Point)

(hWhole :
  IsRectangleContainedBy Geo
    U0 U1 U2 U3 B C M C)

(hProduct :
  IsRectangleContainedBy Geo
    V0 V1 V2 V3 B M M C) :
HilbertScissorsEq Geo
  (hilbertParallelogramTerm Geo U0 U1 U2 U3)
  (hilbertParallelogramTerm Geo V0 V1 V2 V3 +
    hilbertParallelogramTerm Geo M C F G) := by

```

Several features of this signature are mathematically significant.

First, the theorem assumes a configured outer rectangle and its cut. It does not construct that diagram from the bare data $B - M - C$. Euclid performs the square and parallel constructions in his proof; the formal theorem receives the corresponding geometry as explicit hypotheses.

Second, $U_0U_1U_2U_3$ and $V_0V_1V_2V_3$ are arbitrary representatives of

$$\text{Rect}(BC, MC) \quad \text{and} \quad \text{Rect}(BM, MC).$$

The conclusion is therefore not restricted to the literal pieces drawn in the configured cut.

Third, the square $MCFG$ is concrete, but the II.1 representative-independence machinery permits it to stand for the $\text{Rect}(MC, MC)$ piece even when it is not the internal right-hand rectangle of the supplied cut diagram.

Fourth, the conclusion is exactly

```
HilbertScissorsEq
```

rather than the more general relation

```
HilbertScissorsEquicomplementtable
```

and it is not equality of real numbers.

3.8 Relationship with Earlier Book II Propositions

3.8.1 Proposition II.1

II.3 is a direct formal corollary of II.1. The import is

```
import CGJteamLab.Proposition2_1
```

and the only substantial theorem called in the final proof is

```
euclid_proposition_2_1_two
```

The specialization $PQ = MC$ converts the general binary distributive pattern into II.3 immediately.

The relationship can be written schematically as

$$\text{Rect}(BC, PQ) = \text{Rect}(BM, PQ) + \text{Rect}(MC, PQ)$$

followed by

$$PQ := MC.$$

The final term then becomes $\text{Rect}(MC, MC)$, represented by the square on MC .

3.8.2 Proposition II.2

The current Lean file does *not* import Proposition II.2, and its public theorem is not used in the proof. II.2 and II.3 are therefore sibling corollaries of the II.1 binary API rather than a formal chain $\text{II.1} \rightarrow \text{II.2} \rightarrow \text{II.3}$.

The contrast is nevertheless useful. II.2 required cyclic representative rotation because its chosen side order was opposite to the order returned by II.1. II.3 chooses the fixed segment MC in precisely the position expected by II.1, so even that transport disappears. This is evidence that the II.1 API is not merely sufficient but well aligned with the geometry of the next Book II identities.

3.9 Relationship with Book I Infrastructure

Euclid's local proof explicitly calls I.46 to construct the square and I.31 to draw the parallel. The current Lean theorem does not replay those constructions. Instead it assumes the resulting rectangle, cut, parallelograms, side congruence, and square as data.

Below II.1, however, the formal dependency chain still rests on the Book I infrastructure for rectangles, right angles, squares, parallelograms, and exact scissors decomposition. The binary split ultimately uses the exact rectangle cut machinery developed over the Book I scissors layer. Thus the difference is one of modular organization, not a change from synthetic to numerical geometry.

3.10 Formalization Notes and Hidden Geometric Data

The elementary picture hides several obligations that appear explicitly in the current theorem signature:

- $B - M - C$, fixing the cut point and its orientation on the whole side;
- $D - L - E$, locating the opposite endpoint of the cut;
- $D - X - C$ and $M - X - L$, recording the intersection X of the diagonal and the cut line used by the inherited exact scissors theorem;
- `IsParallelogram Geo L D B M` for the left piece;
- `IsParallelogram Geo C E L M` for the right piece;
- $CE \cong MC$, making the outer rectangle have the required fixed second side magnitude;
- a supplied square $MCFG$;
- arbitrary representatives of the whole rectangle and the BM -by- MC product rectangle.

The point X is invisible in Euclid’s verbal statement. It belongs to the triangulated exact-dissection interface inherited from II.1. It should therefore be read as hidden formal geometry rather than as an additional step in Euclid’s classical construction.

There are also useful negative observations. No `SameRay`, `SameSide`, or `OppositeSide` hypothesis occurs in `Proposition2_3.lean` itself, and no side-order swap is performed. Those lower-level obligations have already been absorbed by the reusable rectangle and II.1 layers.

3.11 Geometry, Content, Representation, and Later Arithmetic

3.11.1 Geometry

The geometric layer consists of the rectangle $BCED$, the cut point M , the opposite cut point L , the diagonal witness X , the two parallelogram pieces, the congruence $CE \cong MC$, and the square $MCFG$.

3.11.2 Content / scissors

The conclusion is the exact relation `HilbertScissorsEq`. The public proof does not weaken the result to general equicomplementability. Exactness is inherited from the binary rectangle split and representative transports already proved in II.1.

3.11.3 Representation

Representation is present but simpler than in II.2. The theorem accepts an arbitrary rectangle representative for $\text{Rect}(BC, MC)$ and an arbitrary representative for $\text{Rect}(BM, MC)$, together with a concrete square on MC . The binary II.1 theorem is already independent of the particular concrete rectangles appearing in the cut, so no additional representation transport is required locally.

3.11.4 Later arithmetic / numerical area

Neither layer is used. No segment product is defined, no fixed unit segment is introduced, and no real-valued area function appears. The expression $xz = yz + z^2$ is only the modern algebraic shadow of a geometric scissors identity.

3.12 Axiomatic and Semantic Status

The public theorem assumes

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

and therefore lives in the same Euclidean project layer as II.1 and II.2. Inspection of the current 96-line source shows one public theorem declaration and no proposition-local `axiom`, `sorry`, or `admit`.

This should not be described as “axiom-free.” The result inherits the axioms and typeclass assumptions of the established Hilbert/Euclidean/scissors library transitively. No current `#print axioms` output for II.3 was supplied with the documentation sources, so a transitive axiom list is not asserted here.

The implementation checkpoint records that the II.3 Lean theorem had already been build-checked during formalization. The present documentation pass uses the current source as authoritative but does not contain the full Lean project needed to repeat that build locally.

3.13 Hartshorne’s Semantic Control

Hartshorne’s general analysis of Euclid’s area theory states that the results of Book II are valid when Euclid’s equality of figures is read as equality of content [5]. His condensed list of selected Book II propositions does not separately single out II.3, so the source should not be cited as providing a special proof of this proposition.

The general semantic point is sufficient. II.3 compares a rectangle with the union of another rectangle and a square. In the present reconstruction this comparison is sharpened, for this configured case, to exact scissors equality. The modern identity

$$xz = yz + z^2$$

therefore belongs to Hartshorne’s “algebra of areas” only as an interpretive shadow of operations on geometric figures.

II.3 also clarifies the representative issue. The term called “the square on MC ” need not be the literal internal square-shaped piece of the cut; the formal theorem accepts the separately supplied square $MCFG$ because the rectangle calculus already proves independence of representatives at the scissors level.

3.14 Hilbert’s Layering

Hilbert provides the complementary warning against reading later segment arithmetic into the proof. His synthetic theory of plane content distinguishes finite equidecomposition from the more general content relations, while his arithmetic of segments is introduced later and requires a fixed unit segment and additional parallel constructions [2].

II.3 remains entirely below that arithmetic boundary. Its formal chain is

$$\begin{array}{c} \text{rectangle and square geometry} \\ \Downarrow \\ \text{exact binary dissection from II.1} \\ \Downarrow \\ \text{representative-independent exact scissors equality.} \end{array}$$

In fact II.3 is even more direct than II.2: no geometric commutativity theorem is needed because the ordered containing segments already agree with the II.1 binary interface.

3.15 Dependency Summary

The actual Lean dependency chain is

$$\begin{array}{c} \text{Hilbert incidence + Euclidean plane layer} \\ \Downarrow \\ \text{Book I right-angle, square, parallelogram, and scissors infrastructure} \\ \Downarrow \\ \text{HilbertRectangle} \\ \text{IsRectangleContainedBy} \\ \text{square_is_rectangle_contained_by_side} \\ \Downarrow \\ \text{Proposition II.1 binary API} \\ \text{euclid_proposition_2_1_two} \\ \text{specialized by } PQ = MC \\ \Downarrow \\ \text{euclid_proposition_2_3.} \end{array}$$

The classical local chain is different:

$$\text{I.46} \longrightarrow \text{I.31} \longrightarrow \text{rectangle cut into rectangle + square} \longrightarrow \text{II.3.}$$

New geometric axiom in II.3:	no
New content axiom in II.3:	no
New proposition-local helper theorem:	no
New reusable helper theorem:	no
New <code>HilbertRectangle</code> theorem introduced by II.3:	no
New Interface / Book Zero / Scissors theorem:	no
Reuses an earlier Book II proposition:	yes, II.1
Imports II.2:	no
Direct side-order transport used:	no
Exact <code>HilbertScissorsEq</code> :	yes
General equicomplementability required:	no
Numerical area:	no
Segment multiplication:	no
Public Lean theorem build:	previously checked; not re-run here
Full relevant Lean build:	previously checked; not re-run here
Proposition-local <code>sorry/admit</code> :	no

3.16 Conclusion

Proposition II.3 is one of the cleanest tests of the Book II rectangle API. Classically, Euclid constructs a square on one segment and enlarges it by a parallel construction until the whole rectangle contained by the original line and that segment is obtained. Formally, the same identity is already contained in the binary theorem of II.1.

The decisive specialization is simply

$$PQ = MC.$$

It turns the second part rectangle of II.1 into $\text{Rect}(MC, MC)$, and the supplied square on MC provides exactly that representative. Nothing else is needed.

This also explains the precise relation between II.2 and II.3. Both are formal corollaries of II.1, but II.2 needs geometric side-order transport while II.3 does not. The result is therefore a very short Lean proof without being an algebraic proof in the modern numerical sense: the content remains rectangles, squares, congruent sides, and exact scissors equality throughout.

Chapter 4

Proposition II.4

4.1 Introduction

Proposition II.4 is the first proposition of Book II in the present reconstruction whose formal proof is naturally expressed as a composition of *two different earlier Book II propositions*. The classical theorem says that when a segment is cut into two parts, the square on the whole is equal to the two squares on the parts together with twice the rectangle contained by the parts.

The modern algebraic shadow is the familiar identity

$$(y + z)^2 = y^2 + z^2 + 2yz.$$

The formal proof, however, does not define a product of segment lengths and does not evaluate a numerical area. It remains entirely inside the rectangle and exact-scissors layer established at the beginning of Book II.

The current Lean source begins with

```
import CGJteamLab.Proposition2_2
import CGJteamLab.Proposition2_3
```

and the proof uses those imports exactly as the architecture suggests:

1. Proposition II.2 splits the square on the whole segment into two rectangles pairing the whole with the two parts;
2. Proposition II.3 expands each of those rectangles into one square and one cross rectangle;
3. representation transport identifies the two cross rectangles as two representatives of the same “rectangle contained by” the two parts;
4. addition and transitivity in `HilbertScissorsEq` combine the results.

Thus the substantial new fact at II.4 is not a new geometric construction or a new arithmetic law. It is the observation that the Book II rectangle API is already compositional enough to express the square-of-a-sum pattern using previously proved exact scissors theorems.

4.2 Euclid's Statement and Classical Configuration

4.2.1 The statement

In Heath's wording, Proposition II.4 states:

If a straight line be cut at random, the square on the whole is equal to the squares on the segments and twice the rectangle contained by the segments.

Let AB be cut at C . In the geometric shorthand used throughout this volume, the statement is

$$\text{Sq}(AB) = \text{Sq}(AC) + \text{Sq}(CB) + 2 \text{Rect}(AC, CB).$$

Here Sq and Rect are expository names for geometric figures and their content. The symbol 2 means that two equal rectangle contents occur in the decomposition; it is not scalar multiplication inside the Lean segment language.

4.2.2 The classical diagram

Euclid describes the square $ADEB$ on AB by I.46 and joins the diagonal BD . Through the cut point C he draws CF parallel to the vertical sides of the square, and through the intersection G of CF with BD he draws HK parallel to AB and DE by I.31 [1]. The square is thereby divided into four subfigures named AG , CK , HF , and GE .

The figure is the classical Euclidean configuration. It is deliberately not presented as the literal Lean input diagram. The formal theorem receives three configured rectangle-cut diagrams inherited from II.2 and II.3, rather than reconstructing this single classical picture point by point.

4.3 Classical Proof

Euclid's proof first identifies the lower-right parallelogram $CGKB$ as a square on CB . Since $CF \parallel AD$, the diagonal BD gives the relevant angle equality by I.29. The square has $BA = AD$, so I.5 gives equal base angles in triangle ADB . Combining these angle equalities and using I.6 yields

$$BC = CG.$$

Opposite sides in the surrounding parallelograms are equal by I.34, so the four sides of $CGKB$ are equal. Right-angle information from the parallels and the square then shows that $CGKB$ is a square on CB [1].

The same reasoning identifies HF as the square on AC . The two remaining subfigures AG and GE are complements about the diagonal of the outer square, and Proposition I.43 gives

$$AG = GE.$$

Moreover AG has adjacent side magnitudes AC and CG , while $CG = CB$. Hence AG is the rectangle contained by AC, CB , and $AG + GE$ is twice that rectangle. Since the four subfigures constitute the whole square $ADEB$, Euclid obtains the required equality [1].

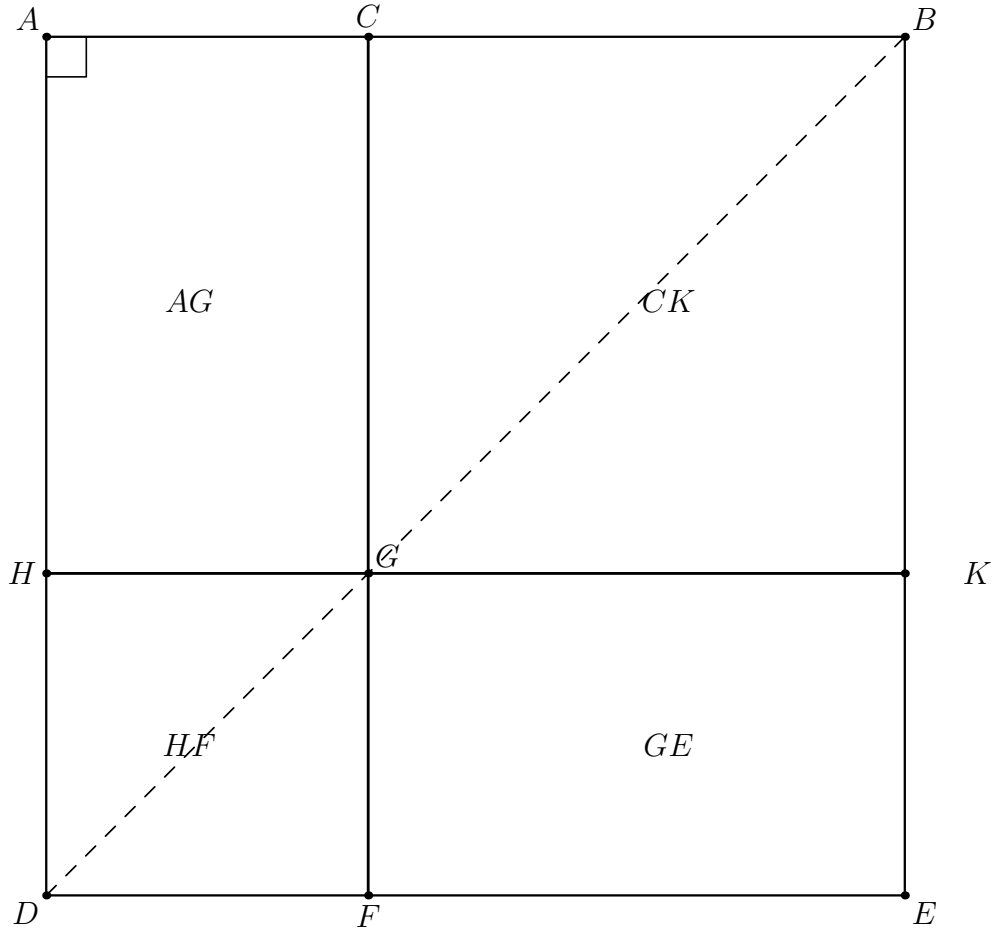


Figure 4.1: The classical configuration of Euclid II.4. The square $ADEB$ is cut at C by CF , the diagonal BD meets it at G , and the parallel HK through G produces four subfigures. Euclid proves that HF and CK are the squares on AC and CB , while the complements AG and GE are equal; AG is the rectangle contained by AC, CB .

The local classical dependency graph is therefore

$$\begin{array}{c}
 \text{I.46: square on } AB \quad + \quad \text{I.31: required parallels} \\
 \Downarrow \\
 \text{I.29} + \text{I.5} + \text{I.6: } BC = CG \\
 \Downarrow \\
 \text{I.34: recognize the two smaller squares} \\
 \Downarrow \\
 \text{I.43: complements } AG = GE \\
 \Downarrow \\
 \text{Sq}(AB) = \text{Sq}(AC) + \text{Sq}(CB) + 2\text{Rect}(AC, CB) \\
 \Downarrow \\
 \text{II.4.}
 \end{array}$$

This classical DAG is important because it is *not* the DAG of the Lean proof. In particular, the formal reconstruction does not locally call I.43 and does not prove the two cross rectangles equal by diagonal complements. That equality has already been absorbed into the representative-independent rectangle calculus developed after Book I.

4.4 The Geometric Identity in the Current Formalization

The public Lean theorem starts with a strict cut

$$A - C - B.$$

It receives a square on AB , two square representatives on the two parts, and two arbitrary representatives of the cross rectangle contained by AC and CB .

At the level of exact scissors terms the final conclusion has the form

$$\text{Sq}(AB) \simeq_{\text{sc}} (\text{Rect}(AC, CB) + \text{Sq}(CA)) + (\text{Rect}(AC, CB) + \text{Sq}(CB)).$$

The use of CA rather than AC in the first square is only an endpoint orientation choice. Segments are unoriented for congruence purposes, so this is the same geometric side magnitude.

The two occurrences of the cross rectangle are deliberately represented by independent quadrilaterals. The theorem therefore says more than “the same literal rectangle occurs twice in one drawing.” It says that any two representatives satisfying

```
IsRectangleContainedBy Geo ... A C C B
```

may occupy the two cross-term positions, because the rectangle layer has already proved representative independence at the exact scissors level.

A modern mnemonic is

$$x = y + z \quad \implies \quad x^2 = y^2 + z^2 + 2yz.$$

This is useful for recognizing the combinatorial pattern only. No object corresponding to yz is constructed by `HilbertSegmentMul` in the II.4 proof.

4.5 Reusable Book II Infrastructure

4.5.1 Proposition II.2: split the square on the whole

The first reusable theorem gives

$$\text{Rect}(AB, AC) + \text{Rect}(AB, CB) \simeq_{\text{sc}} \text{Sq}(AB).$$

In the current source it is called exactly as

```
have hBCA : Geo.Between B C A :=
  (HilbertOrder.between_incidence A C B hACB).2.2.2.2

-----

-- II.2:
--
--   Rect(AB,AC) + Rect(AB,CB) = Square(AB) .
-----

have hII2 :
  HilbertScissorsEq Geo
```

```

(hilbertParallelogramTerm Geo U0 U1 U2 U3 +
  hilbertParallelogramTerm Geo V0 V1 V2 V3)
(hilbertParallelogramTerm Geo A B E0 D0) :=
euclid_proposition_2_2
  Geo
  A B D0 E0 L0 C X0
  hSquareAB
  hACB
  hD0L0E0
  hD0X0B
  hCX0L0
  hII2LeftPar
  hII2RightPar
  U0 U1 U2 U3
  V0 V1 V2 V3
  hAB_AC
  hAB_CB

```

The final II.4 proof uses the symmetry of this relation, so the square becomes the starting point and the two whole-by-part rectangles become the two branches to be expanded.

4.5.2 Proposition II.3: expand a whole-by-part rectangle

Proposition II.3 has the generic geometric pattern

$$\text{Rect}(XY, ZY) \simeq_{\text{sc}} \text{Rect}(XZ, ZY) + \text{Sq}(ZY)$$

for a strict division point Z on XY , with the configured rectangle cut supplied explicitly.

II.4 calls this theorem twice. The right branch uses the original orientation $A - C - B$ directly:

$$\text{Rect}(AB, CB) \simeq_{\text{sc}} \text{Rect}(AC, CB) + \text{Sq}(CB).$$

The left branch reads the same line in the reverse direction $B - C - A$:

$$\text{Rect}(BA, CA) \simeq_{\text{sc}} \text{Rect}(BC, CA) + \text{Sq}(CA).$$

After endpoint reversal and rectangle rotation, its cross term is another representative of $\text{Rect}(AC, CB)$.

4.5.3 Rectangle rotation and endpoint reversal

The only side-order normalization that is mathematically visible in II.4 is

```
rectangle_contained_by_swap
```

applied to the second cross-rectangle representative. It cyclically rotates the concrete rectangle and exchanges the roles of its containing sides while returning an exact scissors certificate for the change of presentation.

Endpoint reversal is handled by the already established congruence theorem

```
CongruentSwapSecond
```

so that AB may be read as BA , AC as CA , and CB as BC when the reversed II.3 invocation requires that orientation. These steps are representation transport, not new metric statements.

4.5.4 Exact scissors addition

Once the two II.3 branches are available, the constructor

```
HilbertScissorsEq.add
```

combines them componentwise. Transitivity then attaches the resulting four-term expression to the square through the symmetric II.2 equality.

No subtraction, cancellation, positivity, content order, or numerical area operation occurs.

4.6 Proof Architecture of the Reconstruction

4.6.1 Step 1: reverse the cut orientation for the left branch

From $A - C - B$ the proof derives $B - C - A$:

```
have hBCA : Geo.Between B C A :=
  (HilbertOrder.between_incidence A C B hACB).2.2.2.2
```

This is the only explicit betweenness reversal performed locally.

4.6.2 Step 2: apply II.2 to the square on the whole

The theorem `euclid_proposition_2_2` produces the exact decomposition of the square into the two rectangles $\text{Rect}(AB, AC)$ and $\text{Rect}(AB, CB)$. Nothing about the smaller squares is used yet.

4.6.3 Step 3: prepare the reversed left II.3 call

The first II.2 rectangle is formally contained by AB, AC , while the reversed II.3 call expects BA, CA . The same concrete rectangle is repackaged using endpoint reversal. The second final cross-rectangle representative is then rotated from the side order AC, CB to CB, AC and read, after endpoint reversal, as BC, CA .

The current source makes these transports explicit:

```
have hAB_AC_rev :
  IsRectangleContainedBy Geo
    U0 U1 U2 U3 B A C A :=
  And.intro
    hAB_AC.1
    (And.intro
      (CongruentSwapSecond
        Geo U0 U1 A B hAB_AC.2.1)
```

```

      (CongruentSwapSecond
        Geo U1 U2 A C hAB_AC.2.2))

-----

-- Rotate the second cross rectangle:
--
--   Rect(AC,CB) -> Rect(CB,AC).
--
-- Endpoint reversal then reads the rotated representative as
-- Rect(BC,CA), exactly what the reversed II.3 needs.
-----

have hSwapResult :=
  rectangle_contained_by_swap
    Geo
    T20 T21 T22 T23
    A C C B
    hCross2

have hCross2Swap :=
  hSwapResult.1

have hRotateCross2 :=
  hSwapResult.2

have hCross2ForLeft :
  IsRectangleContainedBy Geo
    T21 T22 T23 T20 B C C A :=
And.intro
  hCross2Swap.1
  (And.intro
    (CongruentSwapSecond
      Geo T21 T22 C B hCross2Swap.2.1)
    (CongruentSwapSecond
      Geo T22 T23 A C hCross2Swap.2.2))

```

4.6.4 Step 4: expand the left rectangle by II.3

The reversed invocation gives

$$\text{Rect}(BA, CA) \simeq_{\text{sc}} \text{Rect}(BC, CA) + \text{Sq}(CA).$$

The rotated cross term is then transported back to the arbitrary final AC, CB representative. The exact source block is:

```

have hLeftRaw :
  HilbertScissorsEq Geo
    (hilbertParallelogramTerm Geo U0 U1 U2 U3)
    (hilbertParallelogramTerm Geo T21 T22 T23 T20 +
      hilbertParallelogramTerm Geo C A F G) :=
  euclid_proposition_2_3
    Geo
    B A D1 E1 L1 C X1
    hRectLeft
    hAE1_CA

```

```

hBCA
hD1L1E1
hD1X1A
hCX1L1
hLeftLeftPar
hLeftRightPar
F G
hSquareCA
U0 U1 U2 U3
T21 T22 T23 T20
hAB_AC_rev
hCross2ForLeft

-----
-- Return the left cross term to the original arbitrary
-- representative T20-T21-T22-T23.
-----

have hRotateCross2Back :
  HilbertScissorsEq Geo
    (hilbertParallelogramTerm Geo T21 T22 T23 T20)
    (hilbertParallelogramTerm Geo T20 T21 T22 T23) :=
  HilbertScissorsEq.symm
    (Geo := Geo)
  hRotateCross2

have hLeftTarget :
  HilbertScissorsEq Geo
    (hilbertParallelogramTerm Geo U0 U1 U2 U3)
    (hilbertParallelogramTerm Geo T20 T21 T22 T23 +
      hilbertParallelogramTerm Geo C A F G) :=
  HilbertScissorsEq.trans
    (Geo := Geo)
  hLeftRaw
  (HilbertScissorsEq.add
    (Geo := Geo)
    hRotateCross2Back
    (HilbertScissorsEq.refl
      (Geo := Geo)
      (hilbertParallelogramTerm Geo C A F G)))

```

4.6.5 Step 5: expand the right rectangle by II.3

The right branch needs no reversal of the whole segment. It directly produces

$$\text{Rect}(AB, CB) \simeq_{\text{sc}} \text{Rect}(AC, CB) + \text{Sq}(CB).$$

4.6.6 Step 6: add the branches and attach them to the square

The two II.3 relations are added. Finally the result is composed with the symmetric II.2 relation. The closing source is:

```

have hRight :
  HilbertScissorsEq Geo

```

```

      (hilbertParallelogramTerm Geo V0 V1 V2 V3)
      (hilbertParallelogramTerm Geo T10 T11 T12 T13 +
        hilbertParallelogramTerm Geo C B H K) :=
euclid_proposition_2_3
  Geo
  A B D2 E2 L2 C X2
  hRectRight
  hBE2_CB
  hACB
  hD2L2E2
  hD2X2B
  hCX2L2
  hRightLeftPar
  hRightRightPar
  H K
  hSquareCB
  V0 V1 V2 V3
  T10 T11 T12 T13
  hAB_CB
  hCross1

-----
-- Expand both II.2 summands simultaneously.
-----

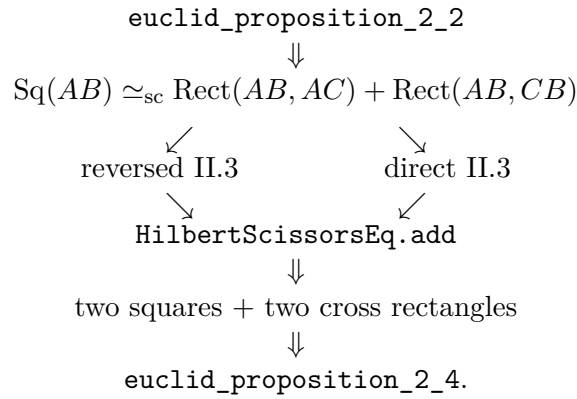
have hExpanded :
  HilbertScissorsEq Geo
    (hilbertParallelogramTerm Geo U0 U1 U2 U3 +
      hilbertParallelogramTerm Geo V0 V1 V2 V3)
    ((hilbertParallelogramTerm Geo T20 T21 T22 T23 +
      hilbertParallelogramTerm Geo C A F G) +
      (hilbertParallelogramTerm Geo T10 T11 T12 T13 +
        hilbertParallelogramTerm Geo C B H K)) :=
  HilbertScissorsEq.add
    (Geo := Geo)
    hLeftTarget
    hRight

-----
-- Square(AB)
--   = Rect(AB,AC) + Rect(AB,CB)
--   = the four terms of II.4.
-----

exact
  HilbertScissorsEq.trans
    (Geo := Geo)
    (HilbertScissorsEq.symm
      (Geo := Geo)
      hII2)
    hExpanded

```

The actual Lean DAG is therefore



This is substantially shorter conceptually than Euclid’s local geometric DAG, but it is not an algebraic shortcut. The earlier propositions already package the relevant geometric dissections as reusable exact-scissors theorems.

4.7 Lean Formalization

The current public theorem signature is copied from the compiling `Proposition2_4.lean` source:

```

theorem euclid_proposition_2_4
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]

  (A B C : Geo.Point)

  -----
  -- II.2 diagram: square on AB cut at C.
  -----

  (D0 E0 L0 X0 : Geo.Point)
  (hSquareAB : IsSquare Geo A B E0 D0)

  (hACB : Geo.Between A C B)
  (hD0LOE0 : Geo.Between D0 L0 E0)
  (hD0X0B : Geo.Between D0 X0 B)
  (hCX0LO : Geo.Between C X0 L0)

  (hII2LeftPar :
    IsParallelogram Geo L0 D0 A C)
  (hII2RightPar :
    IsParallelogram Geo B E0 L0 C)

  -- Arbitrary representatives of Rect(AB,AC) and Rect(AB,CB).
  (U0 U1 U2 U3 : Geo.Point)
  (V0 V1 V2 V3 : Geo.Point)

  (hAB_AC :
    IsRectangleContainedBy Geo
      U0 U1 U2 U3 A B A C)

  (hAB_CB :
```

```

IsRectangleContainedBy Geo
  V0 V1 V2 V3 A B C B)

-----

-- II.3 diagram for the left term.
--
-- We read the divided whole segment in the reverse direction:
--
--   B-C-A.
--
-- Thus II.3 gives
--
--   Rect(BA,CA) = Rect(BC,CA) + Square(CA).
-----

(D1 E1 L1 X1 : Geo.Point)

(hRectLeft :
  IsRectangle Geo B A E1 D1)

(hAE1_CA :
  Geo.Congruent A E1 C A)

(hD1L1E1 : Geo.Between D1 L1 E1)
(hD1X1A : Geo.Between D1 X1 A)
(hCX1L1 : Geo.Between C X1 L1)

(hLeftLeftPar :
  IsParallelogram Geo L1 D1 B C)
(hLeftRightPar :
  IsParallelogram Geo A E1 L1 C)

-- Square on CA, i.e. on the same unoriented segment as AC.
(F G : Geo.Point)
(hSquareCA : IsSquare Geo C A F G)

-----

-- II.3 diagram for the right term:
--
--   Rect(AB,CB) = Rect(AC,CB) + Square(CB).
-----

(D2 E2 L2 X2 : Geo.Point)

(hRectRight :
  IsRectangle Geo A B E2 D2)

(hBE2_CB :
  Geo.Congruent B E2 C B)

(hD2L2E2 : Geo.Between D2 L2 E2)
(hD2X2B : Geo.Between D2 X2 B)
(hCX2L2 : Geo.Between C X2 L2)

(hRightLeftPar :
  IsParallelogram Geo L2 D2 A C)
(hRightRightPar :
  IsParallelogram Geo B E2 L2 C)

```



```

-- Square on CB.
(H K : Geo.Point)
(hSquareCB : IsSquare Geo C B H K)

-----

-- Two arbitrary representatives of the repeated cross rectangle
-- Rect(AC,CB).
-----

(T10 T11 T12 T13 : Geo.Point)
(T20 T21 T22 T23 : Geo.Point)

(hCross1 :
  IsRectangleContainedBy Geo
    T10 T11 T12 T13 A C C B)

(hCross2 :
  IsRectangleContainedBy Geo
    T20 T21 T22 T23 A C C B) :

HilbertScissorsEq Geo
  (hilbertParallelogramTerm Geo A B E0 D0)
  ((hilbertParallelogramTerm Geo T20 T21 T22 T23 +
    hilbertParallelogramTerm Geo C A F G) +
    (hilbertParallelogramTerm Geo T10 T11 T12 T13 +
    hilbertParallelogramTerm Geo C B H K)) := by

```

Several aspects of the signature deserve explicit interpretation.

First, one strict division relation $A - C - B$ is shared by the theorem, but the public statement receives *three* configured cut diagrams:

- the II.2 square-cut diagram, with points D_0, E_0, L_0, X_0 ;
- the left II.3 rectangle-cut diagram, with points D_1, E_1, L_1, X_1 ;
- the right II.3 rectangle-cut diagram, with points D_2, E_2, L_2, X_2 .

The theorem does not construct these configurations from A, C, B alone. Rather, it proves the content identity once the corresponding geometric witnesses have been supplied.

Second, $U_0U_1U_2U_3$ and $V_0V_1V_2V_3$ are arbitrary representatives of the two rectangles produced at the II.2 level. They need not be literal subrectangles of the square ABE_0D_0 .

Third, the cross terms are represented independently by $T_{10}T_{11}T_{12}T_{13}$ and $T_{20}T_{21}T_{22}T_{23}$. Both satisfy the same `IsRectangleContainedBy` predicate for AC, CB . This makes the formal “twice the rectangle” statement representative-independent rather than tied to one duplicated quadrilateral.

Fourth, the conclusion is exactly

```
HilbertScissorsEq Geo ...
```

and not the weaker

```
HilbertScissorsEquicomplementtable Geo ...
```

relation. Exactness is inherited from II.2, II.3, rectangle rotation, formal addition, and transitivity.

4.8 Relationship with Earlier Book II Propositions

4.8.1 Proposition II.1

II.4 does not import `Proposition2_1` directly. Nevertheless II.1 is the underlying distributive core because both II.2 and II.3 are themselves built from its binary rectangle split. The dependency is therefore transitive:

$$\text{II.1} \longrightarrow \text{II.2} \text{ and } \text{II.3} \longrightarrow \text{II.4}.$$

This is the first point where the investment in II.1 becomes visibly compositional at the proposition level. II.2 and II.3 were sibling specializations of the binary II.1 API; II.4 uses both siblings together.

4.8.2 Proposition II.2

II.2 supplies exactly the outer decomposition needed by II.4:

$$\text{Sq}(AB) \simeq_{\text{sc}} \text{Rect}(AB, AC) + \text{Rect}(AB, CB).$$

The current II.4 source imports II.2 explicitly and calls its public theorem once.

4.8.3 Proposition II.3

II.3 supplies the expansion of each whole-by-part rectangle. The current proof calls the same public theorem twice, once after reversing the line orientation and once directly. Thus II.4 is a genuine composition theorem over the public Book II API rather than a restatement of one earlier helper.

4.9 Relationship with Book I Infrastructure

Euclid’s own local proof depends explicitly on I.46, I.31, I.29, I.5, I.6, I.34, and I.43. The formal II.4 theorem does not replay any of those arguments locally. Their geometric content has entered the current project through the established square, parallel, parallelogram, rectangle, and scissors layers and through the already proved II.2 and II.3 theorems.

The contrast with I.43 is especially instructive. Euclid needs I.43 to prove that the two complements AG and GE in one concrete square are equal. Lean instead receives two arbitrary cross-rectangle representatives and normalizes one of them with `rectangle_contained_by_swap`. Representative independence is already built into the rectangle API, so no local complement theorem is required.

The formal proof is therefore not a transcription of Euclid’s diagram. It is a proof over a stronger reusable library whose theorems were themselves proved synthetically from the established Hilbert/Book I geometry.

4.10 Formalization Notes and Hidden Geometric Data

The polished identity hides a substantial amount of explicit positional data. The current signature contains:

- $A - C - B$, fixing the strict order of the original segment;
- a complete square-cut configuration for the II.2 call;
- two complete rectangle-cut configurations for the two II.3 calls;
- three diagonal/cut intersection witnesses X_0, X_1, X_2 ;
- six parallelogram hypotheses certifying the two pieces in each cut;
- one square on the whole and one square on each part;
- two arbitrary II.2 rectangle representatives;
- two arbitrary representatives of the repeated cross rectangle.

The proof then adds two representational obligations not visible in Euclid's statement:

- reverse $A - C - B$ to $B - C - A$ for the left II.3 call;
- rotate one cross-rectangle representative and reverse segment endpoints so that the ordered side data match the reversed II.3 interface.

There is also useful negative information. No `SameRay`, `SameSide`, or `OppositeSide` datum occurs in the public II.4 proof. No new right-angle propagation theorem, no new parallel theorem, and no new scissors constructor is introduced. Those obligations are encapsulated inside the imported rectangle and Book II layers.

4.11 Geometry, Content, Representation, and Later Arithmetic

4.11.1 Geometry

The geometry consists of one segment cut $A - C - B$, one square on the whole, two squares on the parts, and three configured rectangle dissections. The classical source packages these ideas in one square diagram; the Lean theorem packages them as three reusable instances of already formalized cut theorems.

4.11.2 Content / scissors

The conclusion is exact `HilbertScissorsEq`. The proof only uses reflexivity, symmetry, transitivity, compatibility with addition, and exact scissors certificates supplied by II.2, II.3, and rectangle rotation. General equicomplementability is not required.

4.11.3 Representation

Representation is the genuinely nontrivial local layer. Endpoint order on the containing segments must be normalized for the reversed II.3 branch, and one cross rectangle must be cyclically rotated before it can be used in that branch. The final theorem nevertheless permits two completely independent representatives of $\text{Rect}(AC, CB)$.

This is the precise formal content of “twice the rectangle” in the current architecture: two scissors terms representing the same geometric rectangle magnitude occur as separate summands.

4.11.4 Later arithmetic / numerical area

Neither later layer is used. The proof imports no segment multiplication module, chooses no unit segment, proves no distributive law for a product of segment classes, and evaluates no real-valued area function.

The formula

$$(y + z)^2 = y^2 + z^2 + 2yz$$

should therefore remain explicitly labelled as the modern algebraic shadow of the synthetic construction.

4.12 Axiomatic and Semantic Status

The public theorem assumes

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

and therefore belongs to the same Euclidean project layer as II.1–II.3.

The current `Proposition2_4.lean` file introduces no proposition-local `axiom`, `sorry`, or `admit`. The theorem was compiled successfully in the project. The supplied transitive axiom audit is:

```
#print axioms Geometry.euclid_proposition_2_4

'Geometry.euclid_proposition_2_4' depends on axioms:
[propxt, Classical.choice, bookZero_nullSegment1, Quot.sound]
```

Accordingly II.4 should not be described as “axiom-free.” The precise statement is that it introduces no new proposition-specific axiom and no `sorry`; its transitive dependencies are exactly the four declarations listed above. In particular the list contains no `sorryAx`.

The compile result reported for the current theorem is clean. A separate full `lake build` result was not part of the documentation input used here, so the status table below does not claim that stronger check.

4.13 Hartshorne’s Semantic Control

Hartshorne’s Section 22 supplies the appropriate general semantic control for Book II: Euclid’s equalities of plane figures can be interpreted through equal content without first assigning numerical areas [5]. He also explains why the algebraic-looking manipulations of rectangles and squares may be regarded as a kind of geometrical algebra.

II.4 is a particularly vivid example of that interpretation. Its modern shadow is a quadratic identity, but Euclid’s objects are still a large square, two smaller squares, and two rectangles. The current formalization sharpens the configured result from general equal content to exact scissors equality, because every step is built from exact dissection and exact representation transport.

Hartshorne’s “Brief Euclid” explicitly lists II.4 in the form that the square on the whole line equals the squares on its two segments plus twice the rectangle on the two segments [5]. Thus II.4 has a more direct Hartshorne-side textual counterpart than II.2 or II.3. The surrounding Section 22 discussion still supplies the essential semantic interpretation: Euclid’s equality is read as equal content, not as prior equality of numerical areas.

The stronger exact-scissors certificate remains a fact about the present Lean architecture. It should not be attributed to Hartshorne’s general content relation.

4.14 Hilbert’s Layering

Hilbert provides the complementary architectural warning. His plane-content theory and his later arithmetic of segments are distinct layers. Segment multiplication is introduced only later, using a fixed unit segment and parallel constructions, after which commutativity and distributivity become actual arithmetic theorems [2].

II.4 may look like the first place where one should invoke those later laws, because the shadow

$$(y + z)^2 = y^2 + z^2 + 2yz$$

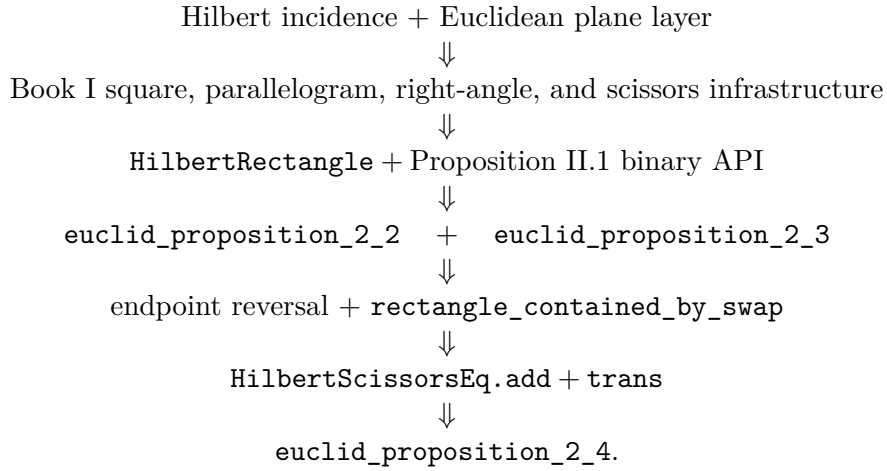
contains both distributivity and commutativity. The compiling source shows that this move is unnecessary. The proof stays below segment arithmetic:

rectangle/square geometry
↓
exact II.2 and II.3 scissors identities
↓
geometric representation transport
↓
exact II.4 scissors identity.

This is a useful architectural checkpoint. Even a full quadratic-looking Book II identity can be derived from the geometric rectangle calculus without crossing into Hilbert’s later segment field.

4.15 Dependency Summary

The actual Lean dependency chain is



The classical local chain is different:

$$\text{I.46} + \text{I.31} + \text{I.29} + \text{I.5} + \text{I.6} + \text{I.34} + \text{I.43} \longrightarrow \text{II.4.}$$

New geometric axiom in II.4:	no
New content axiom in II.4:	no
New proposition-local helper theorem:	no
New reusable helper theorem:	no
New <code>HilbertRectangle</code> theorem introduced by II.4:	no
New Interface / Book Zero / Scissors theorem:	no
Reuses earlier Book II propositions:	yes, II.2 and II.3
Direct import of II.1:	no; inherited through II.2 and II.3
Direct side-order transport used:	yes, <code>rectangle_contained_by_swap</code>
Endpoint-reversal transport used:	yes
Exact <code>HilbertScissorsEq</code> :	yes
General equicomplementability required:	no
Numerical area:	no
Segment multiplication:	no
Public Lean theorem compile:	yes, clean
Full relevant <code>lake build</code> :	not separately reported here
Proposition-local <code>sorry/admit</code> :	no
<code>sorryAx</code> in reported axiom list:	no

4.16 Conclusion

Proposition II.4 is the first strong compositional test of the Book II API. Euclid proves the theorem inside one carefully constructed square: two subfigures are shown to be squares, and I.43 identifies the two remaining complements. The current Lean proof instead works one level higher. It takes II.2 as the outer square decomposition and applies II.3 to each of the two resulting rectangles.

The only local difficulty is representational. One II.3 call uses the reversed order $B - C - A$, and one cross rectangle must be rotated so that its ordered containing sides match that call. Once these transports are made, `HilbertScissorsEq.add` and transitivity close the theorem.

Consequently II.4 requires no new rectangle theorem, no new scissors theorem, no new proposition-local helper, and no segment multiplication. Its apparent quadratic algebra is already present in the synthetic rectangle calculus as a composition of exact geometric decompositions. The reported axiom audit adds no new proposition-specific debt and contains no `sorryAx`.

Appendix A

Greenberg: Foundational and Semantic Control

A.1 Purpose and scope

This appendix records the Greenberg-side source audit for Euclid Book II at the current checkpoint, Propositions II.1–II.3. As in Book I, the purpose is not to force a one-to-one correspondence between Euclid’s proposition numbering and Greenberg’s organization. The relevant question is which parts of the formal architecture Greenberg controls and where the synthetic content theory must be supplied by another source.

A.2 Greenberg’s role at the beginning of Book II

Greenberg distinguishes numerical area from equal content. He describes the latter through finite dissection and reassembly of polygonal figures but does not develop the full synthetic equal-content calculus; he refers the reader to Hartshorne for that theory [4, 5].

This source boundary, already important in the area propositions of Book I, continues unchanged at the opening of Book II. Greenberg’s Euclidean parallel and parallelogram theory is well suited to the geometric shell of rectangle constructions and cuts. It is not the source for the exact scissors relation used by the Lean theorems.

A.3 Comparison table

Table A.1: Euclid II.1–II.3 in Greenberg’s Hilbert-style framework

Euclid	Greenberg counterpart	Status in Greenberg	Source-audit remarks and relevance to formalization
II.1	Rectangle dissection supported geometrically by Euclidean parallel and parallelogram theory.	No dedicated Greenberg proof of II.1 was located in the current audit. Greenberg supplies the geometric setting and the distinction between equal content and numerical area, but not the project’s exact content calculus.	The formal theorem should therefore be read as Greenberg-compatible geometry plus Hilbert/Hartshorne content semantics plus Lean scissors infrastructure. The exact <code>HilbertScissorsEq</code> conclusion is not a theorem attributed to Greenberg.
II.2	Square cut into two rectangles; same Euclidean rectangle/parallelogram shell as II.1.	No separate Greenberg counterpart was located. The source-level role remains the same as for II.1.	Greenberg controls the geometry of the configured square and parallel cut, but the side-order transport and exact scissors equality belong to the formal rectangle/content layer.
II.3	Rectangle and square configuration assembled from the same parallel, parallelogram, and right-angle geometry.	No separate Greenberg counterpart was located. II.3 introduces no new Greenberg-side semantic principle beyond the boundary already identified at II.1.	The formal reduction of II.3 to the binary II.1 theorem is a library architecture fact. Greenberg remains a geometry control source rather than a proof source for the content identity.

A.4 Structural observations from the audit

Three observations should be retained.

- The Euclidean geometry needed to construct rectangles and parallel cuts is naturally expressible in Greenberg’s Hilbert-style framework.
- Equality of content is a separate semantic layer. Greenberg’s decision not to develop its full synthetic calculus is a source boundary, not a defect to be repaired by importing numerical area formulas.
- The formal project’s exact scissors and representative-independence theorems should therefore be documented as project infrastructure, not as Greenberg results.

A.5 II.1–II.3: one stable geometry/content split

Nothing in II.2 or II.3 changes Greenberg’s role established at II.1. All three propositions use elementary Euclidean rectangle geometry, while their conclusions belong to the separate content calculus. The formal distinctions between direct binary splitting, side-order rotation, and square-as-rectangle packaging occur inside the Lean API and do not create three different Greenberg source layers.

This is precisely why the appendix should grow by structural blocks rather than by one narrative section per proposition.

A.6 Result of the Greenberg audit

At the II.3 checkpoint Greenberg remains a foundational and geometric control source. He supports the distinction

synthetic content $\neq$ numerical area

and the Euclidean geometry of rectangles, parallels, and parallelograms. The exact finite-dissection calculus used by the public Lean theorems belongs to the formal development and to the Hilbert/Hartshorne content side of the source hierarchy.

A.7 Current checkpoint

This appendix now concerns Euclid Book II, Propositions II.1–II.3. No new Greenberg-specific boundary appears between these propositions. Later entries should extend the comparison table proposition by proposition, while new narrative sections should be added only when a genuinely new construction, axiomatic, or semantic block appears.

Appendix B

Hartshorne: Equal Content and the Algebra of Areas

B.1 Purpose and scope

This appendix records the Hartshorne-side source audit for Euclid Book II at the current checkpoint, Propositions II.1–II.4. Its purpose is not to create a second proposition-by-proposition commentary. Instead, it records what Hartshorne contributes to the semantic interpretation of Euclid’s rectangle and square equalities, which propositions he singles out, and which source-level distinctions should remain stable as Book II grows.

Hartshorne is especially important here because he separates Euclid’s equality of plane figures from numerical area. In Section 22 he defines equal content synthetically and states that the propositions of Book II are valid when Euclid’s equality of figures is read in that sense [5].

B.2 Hartshorne’s organization of Book II

Hartshorne’s treatment places Book II inside the theory of equal content rather than inside a prior arithmetic of numerical lengths. A rectangle described by two segments can therefore play a product-like geometric role without those segments first becoming numbers. This is the background of the familiar interpretation of Book II as an “algebra of areas” [5].

The phrase is structural, not a claim that Euclid has introduced a field of segment lengths. Modern formulas such as

$$a(b + c) = ab + ac$$

are useful mnemonics for the pattern of a geometric decomposition, but the objects compared by Euclid remain plane figures.

Hartshorne’s brief Euclid summary explicitly lists II.1 and II.4, but does not list II.2 or II.3 among its selected Book II entries. In particular, II.4 is recorded as the statement that the square on the whole line equals the squares on its two segments plus twice the rectangle on the two segments [5]. The omission of II.2 and II.3 should not be read as a negative judgment about them: his general area discussion states that the results of Book II are valid for equal content.

B.3 Comparison table

Table B.1: Euclid II.1–II.4 in Hartshorne’s equal-content framework

Euclid	Hartshorne counterpart	Status in Hartshorne	Source-audit remarks and relevance to formalization
II.1	The rectangle contained by two lines is the sum of the rectangles contained by one of them and the segments of the other.	Hartshorne lists II.1 explicitly in his brief Euclid summary and, more generally, places Book II inside the theory of equal content.	This is the cleanest source match for the formal rectangle-distribution API. The Lean reconstruction strengthens this direct-dissection instance to exact <code>HilbertScissorsEq</code> , but that strengthening is a project-level fact and should not be attributed to Hartshorne’s general equal-content relation.
II.2	Square on the whole line decomposed into the two rectangles pairing the whole with the two parts.	II.2 is not separately listed in Hartshorne’s selected brief Book II summary; it is covered by his general statement that the Book II equalities are valid for equal content.	The source supports the synthetic content reading, while the Lean proof exposes an additional representation issue: the two orders of the containing segments are reconciled geometrically by rectangle rotation rather than by numerical commutativity.
II.3	Rectangle on the whole and one part decomposed into the rectangle on the two parts together with the square on that part.	II.3 is likewise not isolated in the selected brief summary; its semantic status is supplied by Hartshorne’s general Book II equal-content result.	The formal theorem is an exact-scissors specialization of the binary II.1 API. The square is used as a representative of the rectangle contained by the same segment twice. Unlike II.2, no side-order transport is required.
II.4	Square on the whole decomposed into the two squares on the parts together with twice the rectangle contained by the parts.	Hartshorne explicitly lists II.4 in his brief Euclid summary, and Section 22 supplies the equal-content semantics for the equality of the figures.	The Lean theorem strengthens the configured case to exact <code>HilbertScissorsEq</code> . It composes II.2 with two II.3 instances and treats “twice the rectangle” as two independent representatives of the same rectangle content, without numerical area or segment multiplication.

B.4 Semantic observations retained from Hartshorne

Several points should remain stable as later propositions are added.

- Euclid’s equality of plane figures in the area block should not be silently replaced by equality of real-valued areas.
- The phrase “rectangle contained by” two segments names a geometric figure through its adjacent side magnitudes; it is not itself a numerical multiplication operation.
- Hartshorne’s equal content is the appropriate general semantic control for Book II. Exact finite equidecomposability is a stronger local conclusion and must be justified proposition by proposition.

- Modern algebraic identities are useful documentary shadows of the synthetic configurations, not substitutes for the geometric proof.
- Hartshorne’s discussion of property (Z) belongs to the converse direction in the theory of content. Nothing in II.1–II.4 requires recovering metric data from a hypothesis of equal content.

B.5 II.1–II.4: from distribution to composition

The first four propositions form a coherent opening block rather than four independent source problems.

II.1 supplies the general additive mechanism: one side of a rectangle is held fixed while the other is divided. II.2 specializes the same pattern to a square on the whole line, but the formal representation requires exchanging the order of the two containing sides. II.3 is an even more direct binary specialization: fixing the chosen part as the second side gives the required rectangle plus the square on that part without any commutativity transport. II.4 then composes these established patterns: II.2 supplies the two outer rectangles and II.3 expands each one into a square and a cross rectangle.

Thus the modern shadows

$$a(b + c) = ab + ac, \quad x^2 = xy + xz, \quad xz = yz + z^2, \quad (y + z)^2 = y^2 + z^2 + 2yz$$

should be read as four views of one geometric additive mechanism. The formal project keeps this below numerical area and below segment multiplication.

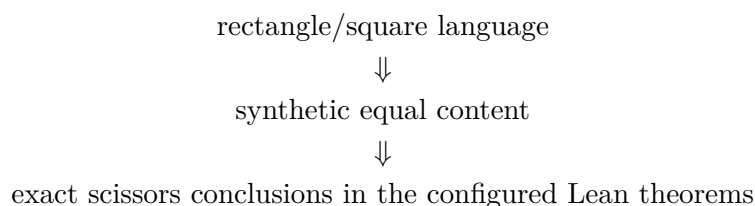
B.6 Result of the Hartshorne audit

At the II.4 checkpoint Hartshorne supplies the main semantic control for the opening of Book II. His contribution is not a separate proof script for each of II.1–II.4, but a framework in which their equalities are understood as equalities of content between plane figures.

The Lean development refines this framework in two ways that are specific to the current implementation. First, the configured II.1–II.4 theorems land at exact `HilbertScissorsEq`. Second, the formal predicate `IsRectangleContainedBy` makes representative independence and side order explicit. These refinements should remain distinguished from claims made by Hartshorne himself.

B.7 Current checkpoint

This appendix now concerns Euclid Book II, Propositions II.1–II.4. The structural picture is



Within the Lean rectangle calculus, II.1 is the reusable distributive core, II.2 tests geometric side-order symmetry, II.3 returns to direct specialization of the binary II.1 theorem, and II.4 demonstrates that those public propositions compose without entering segment arithmetic. Later Hartshorne observations should extend this record by new blocks or new semantic boundaries rather than by duplicating a separate narrative section for every proposition.

Appendix C

Hilbert: Content and Segment Arithmetic

C.1 Purpose and scope

This appendix records the Hilbert-side source audit for Euclid Book II at the current checkpoint, Propositions II.1–II.4. Hilbert is relevant in two distinct ways: Chapter IV supplies the synthetic theory of plane content, whereas Section 15 later develops an arithmetic of segments. The documentary purpose is to keep those layers separate and to identify which one is actually used by the current formal reconstruction.

C.2 Hilbert’s organization of the relevant theory

Hilbert’s Chapter IV distinguishes finite equidecomposability from the more general relation of equicomplementability. The theory of plane area begins on the basis of Groups I–IV and does not need to be introduced as a numerical area function [2].

For the present project the useful hierarchy is

exact finite dissection $\implies$ equicomplementability $\implies$ later numerical area semantics,

with no converse inserted merely for convenience.

Section 15 is a different layer. Hilbert there defines segment addition and a geometric product, using a fixed unit segment and parallel construction, and then proves the familiar algebraic laws, including commutativity and distributivity [2]. That later arithmetic is a valuable control source precisely because it shows what would be required if Book II were to be formalized through segment multiplication. The current Lean development does not take that route.

C.3 Comparison table

Table C.1: Euclid II.1–II.4 against Hilbert’s content and segment-arithmetic layers

Euclid	Hilbert counterpart	Status in the Grundlagen	Source-audit remarks and relevance to formalization
II.1	Chapter IV finite decomposition and content theory; later Section 15 distributivity of segment multiplication is a structural analogue.	No one-to-one numbered Hilbert theorem reproducing Euclid II.1 was located in the current audit. The relevant material is organized by content theory and, later, by segment arithmetic rather than by Euclid’s Book II numbering.	The formal proof belongs to the earlier content/dissection layer. Additivity is obtained from an actual rectangle cut and representative transport, not by invoking the later distributive law for a segment product.
II.2	Chapter IV content plus the geometric symmetry of a rectangle; Section 15 commutativity is only a later arithmetic analogue.	Again the source correspondence is structural rather than proposition-level. Hilbert’s later commutative segment multiplication must not be read backward into the scissors proof.	The Lean theorem makes the distinction concrete: the required exchange of the two containing sides is implemented by cyclic rotation of a rectangle representative, preserving exact scissors content.
II.3	Chapter IV content/dissection; later Section 15 distributivity gives the arithmetic pattern only after segment multiplication has been defined.	No separate proposition-level Hilbert counterpart was located. The source supports the distinction between geometric content operations and the later arithmetic of segments.	II.3 is a direct specialization of the binary II.1 cut with the chosen segment used as the fixed side. The square is packaged as a rectangle contained by the same segment twice. No segment product and no side-order transport are needed.
II.4	Chapter IV finite content composition; later Section 15 distributivity and commutativity give an arithmetic analogue of the square expansion.	No one-to-one Hilbert proposition corresponding to Euclid II.4 was located in the current audit. Hilbert’s source role remains architectural.	The formal theorem composes exact II.2 with two exact II.3 instances and a geometric rectangle rotation. Despite the modern shadow $(a + b)^2 = a^2 + b^2 + 2ab$, no segment product, unit segment, or numerical area is used.

C.4 Structural observations retained from Hilbert

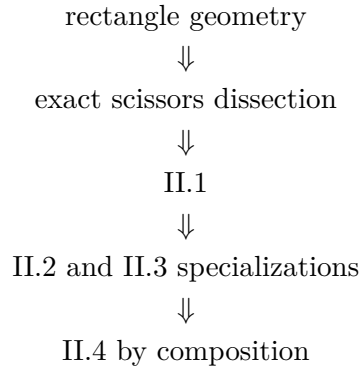
The following distinctions should be preserved as Book II grows.

- Plane content and segment arithmetic are separate theories in the Grundlagen and should remain separate in the documentation.
- Exact finite dissection is stronger than the general content relation; when Lean proves `HilbertScissorsEq`, that fact comes from the local proof architecture rather than from a blanket identification of all equal-content figures with equidecomposable ones.
- Segment multiplication is not primitive. It requires a unit segment, proportion geometry, and parallel constructions before algebraic laws such as commutativity and distributivity can even be stated.
- Consequently, a distributive-looking Euclidean proposition can be formalized below the arithmetic layer when its geometry is literally a finite cut-and-paste argument.

C.5 II.1–II.4: exact dissection below segment arithmetic

The opening four propositions give a useful stress test of the separation. II.1 establishes a reusable exact rectangle split. II.2 shows that the apparent exchange of factors can still be implemented geometrically by rotating a rectangle presentation. II.3 then shows that even this extra symmetry operation is unnecessary when the ordered side data already match the binary II.1 interface. II.4 is the first compositional step: it combines II.2 with two II.3 expansions and uses only exact scissors addition and representation transport.

The resulting formal chain is therefore



not

segment field $\longrightarrow$ multiplication $\longrightarrow$ algebraic simplification.

C.6 Result of the Grundlagen audit

At the II.4 checkpoint Hilbert’s main contribution is architectural. Chapter IV validates the use of a synthetic content layer independently of numerical area, while Section 15 marks a later arithmetic boundary. The current Book II formalization remains on the earlier side of that boundary.

The rectangle API therefore has a clear foundational interpretation: congruent segments and right angles determine representative rectangles, and the scissors layer compares their content. The algebra-like behavior observed in II.1–II.3 is derived from that geometry.

C.7 Current checkpoint

This appendix now concerns Euclid Book II, Propositions II.1–II.4. At this stage the durable Hilbert-side picture is

Groups I–IV geometry $\longrightarrow$ plane-content/dissection layer $\longrightarrow$ rectangle calculus for II.1–II.4,

while Section 15 segment arithmetic remains a later comparison layer rather than a dependency of the proofs.

Appendix D

Borsuk–Szmielw: Axiomatic Layers and Representation

D.1 Purpose and scope

This appendix records the Borsuk–Szmielw-side source audit for Euclid Book II at the current checkpoint, Propositions II.1–II.3. The source is used primarily to control the axiomatic layering of incidence, order, congruence, polygons, parallels, and parallelograms. The question at the beginning of Book II is how far that geometry reaches before a separate theory of content is needed.

D.2 The relevant axiomatic boundary

Borsuk–Szmielw develop rigorous polygonal and Euclidean parallelogram infrastructure, but the current audit has not located a synthetic Euclidean theory of equal content or equicomplementability parallel to Hilbert Chapter IV or Hartshorne Section 22 [3].

That absence is structurally useful. It confirms that one can separate the representation and Euclidean geometry of rectangles from the content relation used to compare them. A rectangle is a special parallelogram, and its subdivision is ordinary polygonal geometry; the assertion that the whole and the pieces have the required common content belongs to a different layer.

D.3 Comparison table

Table D.1: Euclid II.1–II.3 against the Borsuk–Szmielw axiomatic layers

Euclid	Borsuk–Szmielw counterpart	Status / layer	Structural significance for the reconstruction
II.1	Rectangle as a special parallelogram together with polygonal subdivision.	Geometry available; no direct content counterpart located.	The geometric configuration is natural in the source architecture, but the conclusion that the whole rectangle equals the sum of the subrectangles belongs to the separate scissors/content layer of the formal project.

Continued on next page

Table D.1 – continued from previous page

Euclid	Borsuk–Szmielw counterpart	Status / layer	Structural significance for the reconstruction
II.2	Square/rectangle geometry and subdivision by a parallel through the division point.	Geometry available; no direct content counterpart located.	The source can support the representation of the square and its two rectangular parts. Exact scissors equality and the formal side-order transport are not supplied by the audited Borsuk–Szmielw theory.
II.3	Rectangle plus square configuration built from the same Euclidean parallelogram/right-angle infrastructure.	Geometry available; no direct content counterpart located.	No new Borsuk–Szmielw layer appears. The reduction to binary II.1 and the identification of a square with a rectangle contained by the same segment twice belong to the formal rectangle/content API.

D.4 Structural observations retained from the audit

The beginning of Book II reinforces the following separation:

polygon/rectangle geometry + scissors/content relation + representative independence.

Borsuk–Szmielw are strongest on the first term. The second and third are additional layers of the present formal development. This prevents the source audit from silently attributing to the book a theory of equal content that the current audit has not located there.

D.5 II.1–II.3: representation is stable, content is external

Across II.1–II.3 the source role is stable. Rectangle, square, parallel, and parallelogram geometry remain within the Euclidean axiomatic layer. What changes from proposition to proposition is the way the formal scissors API packages and reuses that geometry: general split in II.1, side-order transport in II.2, and direct binary specialization in II.3.

These are important formal distinctions, but they are not reasons to invent three different Borsuk–Szmielw counterparts. The proper source conclusion is that the same geometry/content boundary persists throughout the opening block.

D.6 Consequences for the formal library

The audit supports a modular architecture in which

- Euclidean incidence, parallelism, right-angle, and parallelogram facts live below the Book II content layer;
- rectangle representatives are built from that geometry;
- exact scissors comparison and representative independence are supplied separately;
- later Book II propositions may reuse the rectangle/content API without rebuilding the underlying Euclidean geometry.

D.7 Current checkpoint

This appendix now concerns Euclid Book II, Propositions II.1–II.3. No direct Borsuk–Szmielw content counterpart has been located for the block. The source continues to function as a geometry-and-representation control source, while the equality-of-content semantics is supplied elsewhere.

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